

On some equations concerning the Ramanujan unpublished papers and four-dimensional pseudo-particle. New possible mathematical connections with various parameters of Theoretical Physics, String Theory, Supersymmetry Breaking and Number Theory. II

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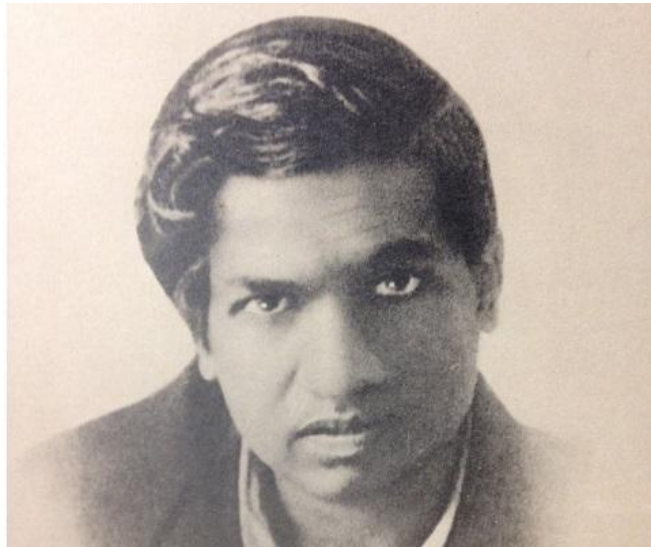
Abstract

In this research thesis (part II), we analyze some equations concerning Ramanujan unpublished papers and four-dimensional pseudo-particle. New possible mathematical connections with various parameters of Theoretical Physics, String Theory, Supersymmetry Breaking and Number Theory.

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Srinivasa Ramanujan (1887-1920)



<https://www.moduscc.it/ramanujan-il-grande-matematico-indiano-13453-131115/>

Vesuvius landscape with gorse – Naples



<https://www.pinterest.it/pin/95068242114589901/>

From:

A. A. Karatsuba, “On the zeros of arithmetic Dirichlet series without Euler product,” *Izv. Ross. Akad. Nauk, Ser. Mat.* 57 (5), 3–14 (1993)

On the Zeros of the Davenport Heilbronn Function

S. A. Gritsenko - Received May 15, 2016 - ISSN 0081-5438, Proceedings of the Steklov Institute of Mathematics, 2017, Vol. 296, pp. 65–87.

We have:

Let

$$\varkappa = \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

and χ_1 be a character modulo 5 such that $\chi_1(2) = i$.

The Davenport–Heilbronn function $f(s)$ is defined by the equality

$$f(s) = \frac{1 - i\varkappa}{2} L(s, \chi_1) + \frac{1 + i\varkappa}{2} L(s, \bar{\chi}_1), \quad \text{where} \quad L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

The function $f(s)$ satisfies the Riemann-type functional equation

$$g(s) = g(1 - s), \quad \text{where} \quad g(s) = \left(\frac{\pi}{5}\right)^{-s/2} \Gamma\left(\frac{s+1}{2}\right) f(s),$$

but there is no Euler product for it.

$$(\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1) = \kappa$$

Input:

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

Decimal approximation:

0.2840790438404122960282918323931261690910880884457375827591626661

...

0.28407904384.... = κ

Alternate forms:

$$\frac{1}{4} \left(\sqrt{10 - 2\sqrt{5}} - 2\sqrt{5} + \sqrt{5(10 - 2\sqrt{5})} - 2 \right)$$

$$\frac{1}{4} (1 + \sqrt{5}) \left(\sqrt{10 - 2\sqrt{5}} - 2 \right)$$

$$\frac{1}{2} \left(-1 - \sqrt{5} + \sqrt{2(5 + \sqrt{5})} \right)$$

Minimal polynomial:

$$x^4 + 2x^3 - 6x^2 - 2x + 1$$

Expanded forms:

$$\frac{\sqrt{10 - 2\sqrt{5}}}{\sqrt{5} - 1} - \frac{2}{\sqrt{5} - 1}$$

$$\frac{1}{4} \sqrt{10 - 2\sqrt{5}} + \frac{1}{4} \sqrt{5(10 - 2\sqrt{5})} + \frac{1}{2} (-1 - \sqrt{5})$$

For $((((\sqrt{10-2\sqrt{5}}-2))(\sqrt{5}-1)))) = 8\pi G$; $G = 0.011303146014$

Indeed:

$$(((\sqrt{10-2\sqrt{5}}-2))(\sqrt{5}-1)))/(8\pi)$$

Input:

$$\frac{\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}}{8\pi}$$

Result:

$$\frac{\sqrt{10-2\sqrt{5}}-2}{8(\sqrt{5}-1)\pi}$$

Decimal approximation:

0.0113031460140052147973750129442035744685760313920017808594909667
...

0.01130314.... = g (gravitational coupling constant)

Property:

$$\frac{-2 + \sqrt{10-2\sqrt{5}}}{8(-1+\sqrt{5})\pi} \text{ is a transcendental number}$$

Alternate forms:

$$\frac{\sqrt{10-2\sqrt{5}}-2\sqrt{5}+\sqrt{5(10-2\sqrt{5})}-2}{32\pi}$$

$$-\frac{1 + \sqrt{5} - \sqrt{2(5 + \sqrt{5})}}{16\pi}$$

$$\frac{-1 - \sqrt{5} + \sqrt{2(5 + \sqrt{5})}}{16\pi}$$

Expanded forms:

$$-\frac{1}{16\pi} - \frac{\sqrt{5}}{16\pi} + \frac{\sqrt{10 - 2\sqrt{5}}}{32\pi} + \frac{\sqrt{5(10 - 2\sqrt{5})}}{32\pi}$$

$$\frac{\sqrt{10 - 2\sqrt{5}}}{8(\sqrt{5} - 1)\pi} - \frac{1}{4(\sqrt{5} - 1)\pi}$$

Series representations:

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{(8\pi)(\sqrt{5} - 1)} = \frac{-2 + \sqrt{9 - 2\sqrt{5}} \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} (9 - 2\sqrt{5})^{-k}}{8\pi \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right)}$$

$$\frac{\sqrt{10 - 2\sqrt{5}} - 2}{(8\pi)(\sqrt{5} - 1)} = \frac{-2 + \sqrt{9 - 2\sqrt{5}} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (9 - 2\sqrt{5})^{-k}}{k!}}{8\pi \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{4}\right)^k \left(-\frac{1}{2}\right)_k}{k!} \right)}$$

$$\frac{\sqrt{10-2\sqrt{5}}-2}{(8\pi)(\sqrt{5}-1)} = \frac{-2+\sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (10-2\sqrt{5}-z_0)^k z_0^{-k}}{k!}}{8\pi \left(-1+\sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (5-z_0)^k z_0^{-k}}{k!}\right)}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

We note that:

$$(((\sqrt{10-2\sqrt{5}}-2))/(\sqrt{5}-1))*((2i(\sqrt{5}-1)t + \sqrt{5}-1)/(2(\sqrt{2(5-\sqrt{5})}-2))) - 2)))$$

Input:

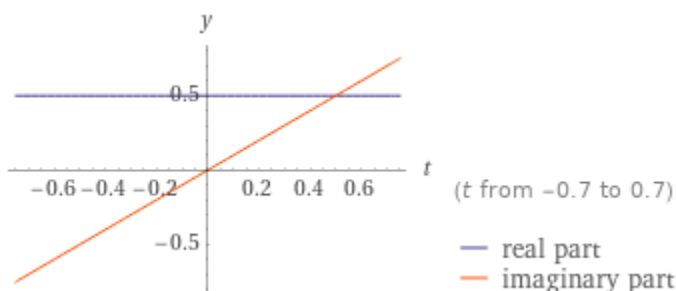
$$\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1} \times \frac{2i(\sqrt{5}-1)t + \sqrt{5}-1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)}$$

i is the imaginary unit

Exact result:

$$\frac{\left(\sqrt{10-2\sqrt{5}}-2\right)\left(2i\left(\sqrt{5}-1\right)t + \sqrt{5}-1\right)}{2\left(\sqrt{5}-1\right)\left(\sqrt{2\left(5-\sqrt{5}\right)}-2\right)}$$

Plot:



Alternate form assuming $t > 0$:

$$\begin{aligned} & \frac{i \sqrt{10 - 2\sqrt{5}} t}{\sqrt{2(5 - \sqrt{5})} - 2} - \frac{2 i t}{\sqrt{2(5 - \sqrt{5})} - 2} + \\ & \frac{\sqrt{5(10 - 2\sqrt{5})}}{2(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} - \frac{\sqrt{10 - 2\sqrt{5}}}{2(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} - \\ & \frac{\sqrt{5}}{(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} + \frac{1}{(\sqrt{5} - 1) \left(\sqrt{2(5 - \sqrt{5})} - 2 \right)} \end{aligned}$$

Alternate forms:

$$\frac{1}{8} (1 + \sqrt{5}) \left(2 i \sqrt{2(3 - \sqrt{5})} t + \sqrt{5} - 1 \right)$$

$$\frac{1}{2} (1 + 2 i t)$$

$$\frac{1}{2} + i t$$

$1/2 + it$ = real part of every nontrivial zero of the Riemann zeta function

Derivative:

$$\frac{d}{dt} \left(\frac{\left(\sqrt{10 - 2\sqrt{5}} - 2 \right) (2 i (\sqrt{5} - 1) t + \sqrt{5} - 1)}{(\sqrt{5} - 1) \left(2 \left(\sqrt{2(5 - \sqrt{5})} - 2 \right) \right)} \right) = i$$

Indefinite integral:

$$\int \frac{(\sqrt{10-2\sqrt{5}}-2)(2i(\sqrt{5}-1)t+\sqrt{5}-1)}{(\sqrt{5}-1)\left(2\left(\sqrt{2(5-\sqrt{5})}-2\right)\right)} dt = \frac{t}{2} + \frac{it^2}{2} + \text{constant}$$

And again:

$$(((\sqrt{10-2\sqrt{5}}-2))/(2x))) * ((2i(\sqrt{5}-1)t + \sqrt{5}-1)/(2(\sqrt{2(5-\sqrt{5})}-2))) = (1/2 + it)$$

Input:

$$\frac{\sqrt{10-2\sqrt{5}}-2}{2x} \times \frac{2i(\sqrt{5}-1)t + \sqrt{5}-1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)} = \frac{1}{2} + it$$

i is the imaginary unit

Exact result:

$$\frac{(\sqrt{10-2\sqrt{5}}-2)(2i(\sqrt{5}-1)t + \sqrt{5}-1)}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} = \frac{1}{2} + it$$

Alternate form assuming t and x are real:

$$\frac{\sqrt{5}-1}{x} = 2$$

Alternate form:

$$\frac{(\sqrt{5} - 1)(1 + 2it)}{4x} = \frac{1}{2} + it$$

Alternate form assuming t and x are positive:

$$2x + 1 = \sqrt{5}$$

Expanded forms:

$$\begin{aligned} & \frac{i\sqrt{5(10-2\sqrt{5})}t}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{i\sqrt{10-2\sqrt{5}}t}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{i\sqrt{5}t}{\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \\ & \frac{it}{\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \frac{\sqrt{5(10-2\sqrt{5})}}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \frac{\sqrt{10-2\sqrt{5}}}{4\left(\sqrt{2(5-\sqrt{5})}-2\right)x} - \\ & \frac{\sqrt{5}}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} + \frac{1}{2\left(\sqrt{2(5-\sqrt{5})}-2\right)x} = \frac{1}{2} + it \end{aligned}$$

$$\frac{i\sqrt{5}t}{2x} - \frac{it}{2x} + \frac{\sqrt{5}}{4x} - \frac{1}{4x} = \frac{1}{2} + it$$

Solutions:

$$t = \frac{i}{2}, \quad x \neq 0$$

$$x = \frac{\sqrt{5}}{2} - \frac{1}{2}$$

Input:

$$\frac{\sqrt{5}}{2} - \frac{1}{2}$$

Decimal approximation:

0.6180339887498948482045868343656381177203091798057628621354486227

...

$$0.6180339887\dots = \frac{1}{\phi}$$

Solution for the variable x:

$$x = \frac{-2i\sqrt{5}t + 2it - \sqrt{5} + 1}{-2 - 4it}$$

Implicit derivatives:

$$\frac{\partial x(t)}{\partial t} = \frac{2(-1 + \sqrt{5} - 2x)x}{(-1 + \sqrt{5})(-i + 2t)}$$

$$\frac{\partial t(x)}{\partial x} = \frac{(-1 + \sqrt{5})(-i + 2t)}{2(-1 + \sqrt{5} - 2x)x}$$

From:

Lost Notebook and Other Unpublished Papers - By Srinivasa Ramanujan -

Publication date: 1920-08-14

$$\frac{\sqrt{x}}{1+\frac{x}{1}+\dots} = \frac{1}{4t} \left\{ (1+t \cdot \frac{\sqrt{5+1}}{2}) \sqrt{1-t} - \sqrt{(1-t)(1+t \cdot \frac{\sqrt{5+1}}{2})^2 - 2t(\sqrt{5+1})} \right. \\ \left. - (1-t \cdot \frac{\sqrt{5-1}}{2}) \sqrt{1-t} + \sqrt{(1-t)(1-t \cdot \frac{\sqrt{5-1}}{2})^2 + 2t(\sqrt{5-1})} \right\}$$

$$\frac{1}{4t} \left\{ (1+t \cdot \frac{\sqrt{5+1}}{2}) \sqrt{1-t} - \sqrt{(1-t)(1+t \cdot \frac{\sqrt{5+1}}{2})^2 - 2t(\sqrt{5+1})} \right. \\ \left. - (1-t \cdot \frac{\sqrt{5-1}}{2}) \sqrt{1-t} + \sqrt{(1-t)(1-t \cdot \frac{\sqrt{5-1}}{2})^2 + 2t(\sqrt{5-1})} \right\}$$

1/(4t)

[(1+t*((sqrt(5)+1)/2) sqrt(1-t)-sqrt((((1-t)(1+t*((sqrt(5)+1)/2))^2-2t(sqrt(5)+1)))))]

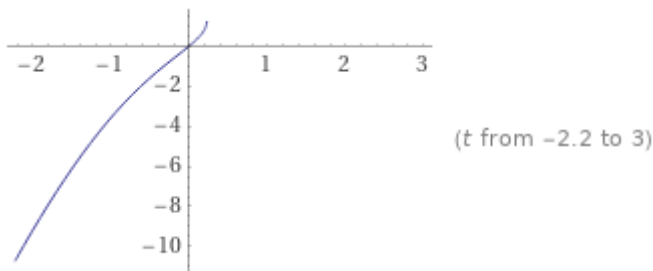
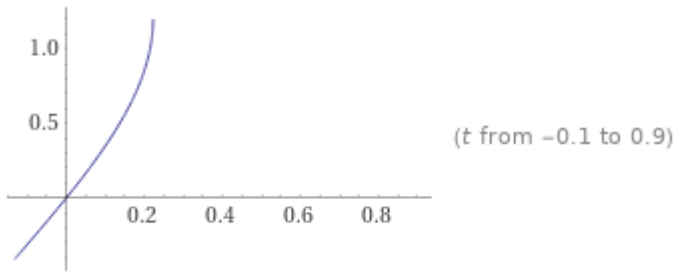
Input

$$\left(1+t \left(\frac{1}{2}(\sqrt{5}+1)\right)\right) \sqrt{1-t} - \sqrt{(1-t) \left(1+t \left(\frac{1}{2}(\sqrt{5}+1)\right)\right)^2 - 2t(\sqrt{5}+1)}$$

Exact result

$$\sqrt{1-t} \left(\frac{1}{2}(1+\sqrt{5})t+1\right) - \sqrt{(1-t) \left(\frac{1}{2}(1+\sqrt{5})t+1\right)^2 - 2(1+\sqrt{5})t}$$

Plots



Alternate forms

$$\frac{1}{2} \sqrt{1-t} (\sqrt{5} t + t + 2) - \sqrt{-\frac{1}{4} (t-1) (\sqrt{5} t + t + 2)^2 - 2(1+\sqrt{5})t}$$

$$\frac{1}{2} \left(-\sqrt{-(1+\sqrt{5})^2 t^3 - (-1-\sqrt{5})(\sqrt{5}-3)t^2 - 4(2+\sqrt{5})t + 4} - \sqrt{1-t} (-(1+\sqrt{5})t - 2) \right)$$

$$\sqrt{1-t} \left(\left(\frac{1}{2} + \frac{\sqrt{5}}{2} \right) t + 1 \right) - \frac{1}{2} \sqrt{-(1+\sqrt{5})^2 t^3 - (-1-\sqrt{5})(\sqrt{5}-3)t^2 - 4(2+\sqrt{5})t + 4}$$

Expanded form

$$\frac{\frac{1}{2}\sqrt{5}\sqrt{1-t}t + \frac{1}{2}\sqrt{1-t}t + \sqrt{1-t}}{\sqrt{(1-t)\left(\frac{1}{2}(1+\sqrt{5})t+1\right)^2 - 2(1+\sqrt{5})t}}$$

Derivative

$$\begin{aligned} \frac{d}{dt} \left(\left(1 + \frac{1}{2}t(\sqrt{5}+1) \right) \sqrt{1-t} - \sqrt{(1-t)\left(1 + \frac{1}{2}t(\sqrt{5}+1)\right)^2 - 2t(\sqrt{5}+1)} \right) = \\ \frac{3(3+\sqrt{5})t^2 + 2(\sqrt{5}-1)t + 2(2+\sqrt{5})}{2\sqrt{-2(3+\sqrt{5})t^3 - 2(\sqrt{5}-1)t^2 - 4(2+\sqrt{5})t + 4}} - \\ \frac{(1+\sqrt{5})t+2}{4\sqrt{1-t}} + \frac{1}{2}(1+\sqrt{5})\sqrt{1-t} \end{aligned}$$

$$[-(1-t*((\sqrt{5}-1)/2)\sqrt{1-t})+\sqrt{((((1-t)(1-t*((\sqrt{5}-1)/2))^2+2t(\sqrt{5}-1))))}]$$

Input

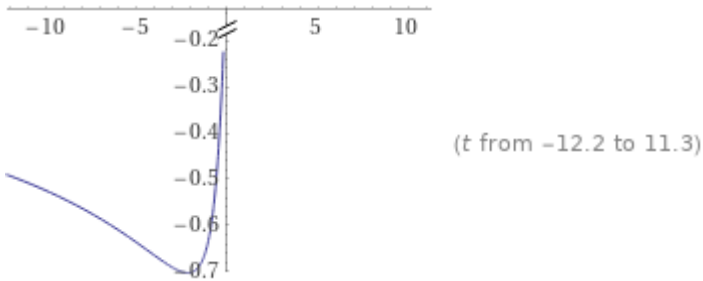
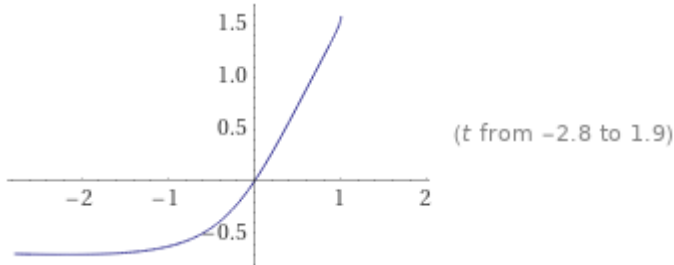
$$-\left(1-t\left(\frac{1}{2}(\sqrt{5}-1)\right)\right)\sqrt{1-t} + \sqrt{(1-t)\left(1-t\left(\frac{1}{2}(\sqrt{5}-1)\right)\right)^2 + 2t(\sqrt{5}-1)}$$

Exact result

$$\sqrt{(1-t)\left(1-\frac{1}{2}(\sqrt{5}-1)t\right)^2 + 2(\sqrt{5}-1)t} - \sqrt{1-t}\left(1-\frac{1}{2}(\sqrt{5}-1)t\right)$$

Plots

(figures that can be related to the open strings)



Alternate forms

$$\frac{1}{2} \sqrt{1-t} ((\sqrt{5}-1)t-2) + \sqrt{2(\sqrt{5}-1)t - \frac{1}{4}(t-1)((\sqrt{5}-1)t-2)^2}$$

$$\sqrt{(1-t) \left(\frac{1}{2} (1-\sqrt{5})t+1 \right)^2 + (2\sqrt{5}-2)t} - \sqrt{1-t} \left(\frac{1}{2} (1-\sqrt{5})t+1 \right)$$

$$\frac{1}{2} \left(\sqrt{-(\sqrt{5}-1)^2 t^3 - (-3-\sqrt{5})(\sqrt{5}-1)t^2 + 4(\sqrt{5}-2)t + 4} + \sqrt{1-t} ((\sqrt{5}-1)t-2) \right)$$

Expanded form

$$\frac{1}{2} \sqrt{5} \sqrt{1-t} t - \frac{1}{2} \sqrt{1-t} t - \sqrt{1-t} + \sqrt{(1-t) \left(1 - \frac{1}{2} (\sqrt{5} - 1) t\right)^2 + 2(\sqrt{5} - 1) t}$$

Derivative

$$\begin{aligned} \frac{d}{dt} \left(- \left(1 - \frac{1}{2} t (\sqrt{5} - 1)\right) \sqrt{1-t} + \sqrt{(1-t) \left(1 - \frac{1}{2} t (\sqrt{5} - 1)\right)^2 + 2 t (\sqrt{5} - 1)} \right) = \\ \frac{1 - \frac{1}{2} (\sqrt{5} - 1) t}{2 \sqrt{1-t}} - \frac{1}{2} (1 - \sqrt{5}) \sqrt{1-t} + \\ \frac{-(1 - \frac{1}{2} (\sqrt{5} - 1) t)^2 + (1 - \sqrt{5}) (1-t) (1 - \frac{1}{2} (\sqrt{5} - 1) t) + 2(\sqrt{5} - 1)}{2 \sqrt{(1-t) \left(1 - \frac{1}{2} (\sqrt{5} - 1) t\right)^2 + 2(\sqrt{5} - 1) t}} \end{aligned}$$

$$\begin{aligned} & 1/(4t) (((1/2 \sqrt{1-t} (\sqrt{5}t+t+2) - \sqrt{-1/4 (t-1)(\sqrt{5}t+t+2)^2 - 2(1+\sqrt{5})t} - \\ & 2(1+\sqrt{5})t))) (((1/2 \sqrt{1-t} ((\sqrt{5}-1)t-2) + \sqrt{2(\sqrt{5}-1)t - 1/4 (t-1)((\sqrt{5}-1)t-2)^2}))) \end{aligned}$$

Input

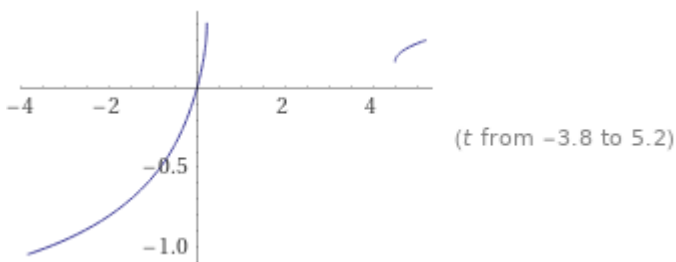
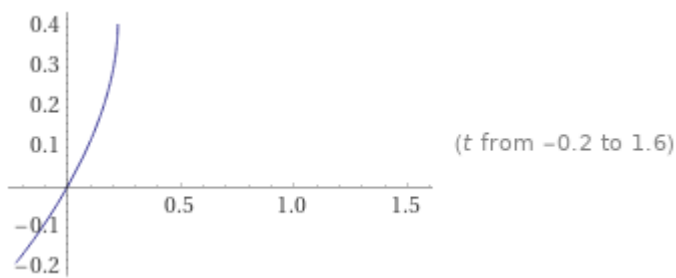
$$\begin{aligned} \frac{1}{4t} \left(\frac{1}{2} \sqrt{1-t} (\sqrt{5} t + t + 2) - \sqrt{-\frac{1}{4} (t-1) (\sqrt{5} t + t + 2)^2 - 2(1+\sqrt{5}) t} \right) \\ \left(\frac{1}{2} \sqrt{1-t} ((\sqrt{5} - 1) t - 2) + \sqrt{2(\sqrt{5} - 1) t - \frac{1}{4} (t-1) ((\sqrt{5} - 1) t - 2)^2} \right) \end{aligned}$$

Exact result

$$\begin{aligned} \frac{1}{4t} \left(\frac{1}{2} \sqrt{1-t} (\sqrt{5} t + t + 2) - \sqrt{-\frac{1}{4} (t-1) (\sqrt{5} t + t + 2)^2 - 2(1+\sqrt{5}) t} \right) \\ \left(\frac{1}{2} \sqrt{1-t} ((\sqrt{5} - 1) t - 2) + \sqrt{2(\sqrt{5} - 1) t - \frac{1}{4} (t-1) ((\sqrt{5} - 1) t - 2)^2} \right) \end{aligned}$$

Plots

(figures that can be related to the open strings)



For $t = 5$:

$$\left(\left(\frac{1}{2} \sqrt{1-5} (\sqrt{5} \times 5 + 5 + 2) - \sqrt{-\frac{1}{4} (5-1) (\sqrt{5} \times 5 + 5 + 2)^2 - 2(1+\sqrt{5}) \times 5} \right) \left(\frac{1}{2} \sqrt{1-5} ((\sqrt{5}-1) \times 5 - 2) + \sqrt{2(\sqrt{5}-1) \times 5 - \frac{1}{4} (5-1) ((\sqrt{5}-1) \times 5 - 2)^2} \right) \right) / (20)$$

Input

$$\frac{1}{20} \left(\left(\frac{1}{2} \sqrt{1-5} (\sqrt{5} \times 5 + 5 + 2) - \sqrt{-\frac{1}{4} (5-1) (\sqrt{5} \times 5 + 5 + 2)^2 - 2(1+\sqrt{5}) \times 5} \right) \left(\frac{1}{2} \sqrt{1-5} ((\sqrt{5}-1) \times 5 - 2) + \sqrt{2(\sqrt{5}-1) \times 5 - \frac{1}{4} (5-1) ((\sqrt{5}-1) \times 5 - 2)^2} \right) \right)$$

Result

$$\frac{1}{20} \left(i(7 + 5\sqrt{5}) - i\sqrt{10(1 + \sqrt{5}) + (7 + 5\sqrt{5})^2} \right) \\ \left(i(5(\sqrt{5} - 1) - 2) + i\sqrt{(5(\sqrt{5} - 1) - 2)^2 - 10(\sqrt{5} - 1)} \right)$$

Decimal approximation

0.2799680455773598942628252730961953733026286914732727396074447211

...

0.2799680455....

Alternate forms

$$\frac{1}{20} \left(2\sqrt{2(23 + 10\sqrt{5})} - 5\sqrt{5} - 7 \right) \left(2\sqrt{2(23 - 10\sqrt{5})} + 5\sqrt{5} - 7 \right)$$

$$\frac{1}{5} \left(-19 + 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})} \right)$$

root of $5x^4 + 76x^3 + 186x^2 - 76x + 5$ near $x = 0.279968$

Minimal polynomial

$$5x^4 + 76x^3 + 186x^2 - 76x + 5$$

Expanded forms

$$-\frac{19}{5} + \frac{2\sqrt{29}}{5} - \frac{7}{20} \sqrt{184 - 80\sqrt{5}} - \frac{7}{20} \sqrt{184 + 80\sqrt{5}} + \frac{1}{4} \left(\sqrt{920 + 400\sqrt{5}} - \sqrt{920 - 400\sqrt{5}} \right)$$

$$-\frac{19}{5} - \frac{7}{20} \sqrt{10(1 + \sqrt{5}) + (7 + 5\sqrt{5})^2} + \frac{1}{4} \sqrt{5(10(1 + \sqrt{5}) + (7 + 5\sqrt{5})^2)} - \frac{7}{20} \sqrt{(5(\sqrt{5} - 1) - 2)^2 - 10(\sqrt{5} - 1)} - \frac{1}{4} \sqrt{5((5(\sqrt{5} - 1) - 2)^2 - 10(\sqrt{5} - 1))} + \frac{1}{20} \sqrt{(10(1 + \sqrt{5}) + (7 + 5\sqrt{5})^2)((5(\sqrt{5} - 1) - 2)^2 - 10(\sqrt{5} - 1))}$$

From which:

$$1 + 1/(2e * (((1/2 \sqrt{1-5}) (\sqrt{5} 5 + 5 + 2) - \sqrt{-1/4 (5-1)(\sqrt{5} 5 + 5 + 2)^2 - 2(1 + \sqrt{5}) 5}) (1/2 \sqrt{1-5}) ((\sqrt{5} - 1) 5 - 2) + \sqrt{2(\sqrt{5} - 1) 5 - 1/4 (5-1)((\sqrt{5} - 1) 5 - 2)^2}))) / (20)))$$

Input

$$1 + 1 / \left(2e \left(\frac{1}{20} \left(\left(\frac{1}{2} \sqrt{1-5} (\sqrt{5} \times 5 + 5 + 2) - \sqrt{-\frac{1}{4} (5-1) (\sqrt{5} \times 5 + 5 + 2)^2 - 2(1 + \sqrt{5}) \times 5} \right) \left(\frac{1}{2} \sqrt{1-5} ((\sqrt{5} - 1) \times 5 - 2) + \sqrt{2(\sqrt{5} - 1) \times 5 - \frac{1}{4} (5-1) ((\sqrt{5} - 1) \times 5 - 2)^2} \right) \right) \right) \right)$$

Exact result

$$1 + 10 / \left(\left(i(7 + 5\sqrt{5}) - i\sqrt{10(1 + \sqrt{5}) + (7 + 5\sqrt{5})^2} \right) \right. \\ \left. \left(i(5(\sqrt{5} - 1) - 2) + i\sqrt{(5(\sqrt{5} - 1) - 2)^2 - 10(\sqrt{5} - 1)} \right) e \right)$$

Decimal approximation

1.6570025525820071233794126422959604046529846942408013875875810439
...

1.657002552582.... result that is very near to the 14th root of the following
Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Property

$$1 + 10 / \left(\left(i(7 + 5\sqrt{5}) - i\sqrt{10(1 + \sqrt{5}) + (7 + 5\sqrt{5})^2} \right) \right. \\ \left. \left(i(-2 + 5(-1 + \sqrt{5})) + i\sqrt{-10(-1 + \sqrt{5}) + (-2 + 5(-1 + \sqrt{5}))^2} \right) e \right)$$

is a transcendental number

Alternate forms

$$1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e}$$

$$1 - \frac{10}{\left(-7 + 5\sqrt{5} + 2\sqrt{46 - 20\sqrt{5}} \right) \left(7 + 5\sqrt{5} - 2\sqrt{46 + 20\sqrt{5}} \right) e}$$

$$1 + \frac{10}{\left(-7 + 5\sqrt{5} + 2\sqrt{2(23 - 10\sqrt{5})}\right)\left(-7 - 5\sqrt{5} + 2\sqrt{2(23 + 10\sqrt{5})}\right)e}$$

Series representations

$$\begin{aligned} & 1 + 1 / \left(\frac{1}{20} (2e) \left(\left(\frac{1}{2} \sqrt{1-5} (\sqrt{5} 5 + 5 + 2) - \right. \right. \right. \\ & \quad \left. \left. \sqrt{-\frac{1}{4} (5-1) (\sqrt{5} 5 + 5 + 2)^2 - 2(1+\sqrt{5}) 5} \right) \right. \\ & \quad \left. \left(\frac{1}{2} \sqrt{1-5} ((\sqrt{5}-1) 5 - 2) + \right. \right. \\ & \quad \left. \left. \sqrt{2(\sqrt{5}-1) 5 - \frac{1}{4} (5-1) ((\sqrt{5}-1) 5 - 2)^2} \right) \right) \Bigg) = \\ & 1 + 10 / \left(e \left(\frac{1}{2} \sqrt{-5} \left(\sum_{k=0}^{\infty} (-5)^{-k} \binom{\frac{1}{2}}{k} \right) \left(-2 + 5 \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right) \right) \right. \right. \\ & \quad \left. \sqrt{-1 - (-2 + 5(-1 + \sqrt{5}))^2 + 10(-1 + \sqrt{5})} \right. \\ & \quad \left. \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} (-1 - (-2 + 5(-1 + \sqrt{5}))^2 + 10(-1 + \sqrt{5}))^{-k} \right) \\ & \quad \left(\frac{1}{2} \sqrt{-5} \left(\sum_{k=0}^{\infty} (-5)^{-k} \binom{\frac{1}{2}}{k} \right) \left(7 + 5\sqrt{4} \sum_{k=0}^{\infty} 4^{-k} \binom{\frac{1}{2}}{k} \right) - \right. \\ & \quad \left. \sqrt{-1 - 10(1 + \sqrt{5}) - (7 + 5\sqrt{5})^2} \right. \\ & \quad \left. \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} (-1 - 10(1 + \sqrt{5}) - (7 + 5\sqrt{5})^2)^{-k} \right) \Bigg) \end{aligned}$$

$$\begin{aligned}
& 1 + 1 / \left(\frac{1}{20} (2e) \left(\left(\frac{1}{2} \sqrt{1-5} (\sqrt{5} 5 + 5 + 2) - \right. \right. \right. \\
& \quad \left. \left. \sqrt{-\frac{1}{4} (5-1) (\sqrt{5} 5 + 5 + 2)^2 - 2(1+\sqrt{5}) 5} \right) \right. \\
& \quad \left. \left(\frac{1}{2} \sqrt{1-5} ((\sqrt{5}-1) 5 - 2) + \right. \right. \\
& \quad \left. \left. \sqrt{2(\sqrt{5}-1) 5 - \frac{1}{4} (5-1) ((\sqrt{5}-1) 5 - 2)^2} \right) \right) \right) = 1 + 10 / \\
& \left(e \left(\frac{1}{2} \sqrt{-5} \left(\sum_{k=0}^{\infty} \frac{(-5)^{-k} (-1)^k \left(-\frac{1}{2}\right)_k}{k!} \right) \left(-2 + 5 \left(-1 + \sqrt{4} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{4}\right)^k \left(-\frac{1}{2}\right)_k}{k!} \right) \right) + \right. \\
& \quad \left. \sqrt{-1 - (-2 + 5(-1 + \sqrt{5}))^2 + 10(-1 + \sqrt{5})} \right. \\
& \quad \left. \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(-1 - (-2 + 5(-1 + \sqrt{5}))^2 + 10(-1 + \sqrt{5})\right)^{-k}}{k!} \right) \\
& \left(\frac{1}{2} \sqrt{-5} \left(\sum_{k=0}^{\infty} \frac{(-5)^{-k} (-1)^k \left(-\frac{1}{2}\right)_k}{k!} \right) \left(7 + 5 \sqrt{4} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{4}\right)^k \left(-\frac{1}{2}\right)_k}{k!} \right) - \right. \\
& \quad \left. \sqrt{-1 - 10(1 + \sqrt{5}) - (7 + 5 \sqrt{5})^2} \right. \\
& \quad \left. \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(-1 - 10(1 + \sqrt{5}) - (7 + 5 \sqrt{5})^2\right)^{-k}}{k!} \right) \right)
\end{aligned}$$

$$\begin{aligned}
& 1 + 1 / \left(\frac{1}{20} (2e) \left(\left(\frac{1}{2} \sqrt{1-5} (\sqrt{5} 5 + 5 + 2) - \right. \right. \right. \\
& \quad \left. \left. \sqrt{-\frac{1}{4} (5-1) (\sqrt{5} 5 + 5 + 2)^2 - 2(1 + \sqrt{5}) 5} \right) \right. \\
& \quad \left. \left(\frac{1}{2} \sqrt{1-5} ((\sqrt{5} - 1) 5 - 2) + \right. \right. \\
& \quad \left. \left. \sqrt{2(\sqrt{5} - 1) 5 - \frac{1}{4} (5-1) ((\sqrt{5} - 1) 5 - 2)^2} \right) \right) \right) = \\
& 1 + 10 / \left(e \left(\frac{1}{2} \sqrt{z_0} \left(\sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (-4 - z_0)^k z_0^{-k}}{k!} \right) \right. \right. \\
& \quad \left. \left(-2 + 5 \left(-1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (5 - z_0)^k z_0^{-k}}{k!} \right) \right) \right) + \\
& \quad \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(-(-2 + 5(-1 + \sqrt{5}))^2 + 10(-1 + \sqrt{5}) - z_0 \right)^k z_0^{-k}}{k!} \\
& \quad \left. \right) \\
& \quad \left(\frac{1}{2} \sqrt{z_0} \left(\sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (-4 - z_0)^k z_0^{-k}}{k!} \right) \right. \\
& \quad \left(7 + 5 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k (5 - z_0)^k z_0^{-k}}{k!} \right) - \\
& \quad \left. \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(-10(1 + \sqrt{5}) - (7 + 5 \sqrt{5})^2 - z_0 \right)^k z_0^{-k}}{k!} \right) \right) \right)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

From:

$$1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e}$$

we obtain:

$$(1 + (19 - 2 \sqrt{29} + \sqrt{2(251 - 38 \sqrt{29})})) / (10e)^{15-233+4\pi + (((\sqrt{10-2\sqrt{5}} - 2)) / (\sqrt{5}-1)))}$$

Input

$$\left(1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e} \right)^{15} - 233 + 4\pi + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}$$

Decimal approximation

1729.0112183865073470934244999279842718987764020656756211830672869

...

1729.0112183....

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

$$2 * (((1/27 * ((1 + (19 - 2 \sqrt{29} + \sqrt{2(251 - 38 \sqrt{29})})) / (10e)^{15-233+4\pi + (((\sqrt{10-2\sqrt{5}} - 2)) / (\sqrt{5}-1))) - 1))^2))$$

Input

$$2 \left(\frac{1}{27} \left(\left(1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e} \right)^{15} - 233 + 4\pi + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} - 1 \right)^2 \right)$$

Exact result

$$\frac{2}{729} \left(-234 + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} + \left(1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e} \right)^{15} + 4\pi \right)^2$$

Decimal approximation

8192.1063672691950271141354984372865519066049043944336486950270290

...

8192.106367.... $\approx$ 8192

The total amplitude vanishes for gauge group SO(8192), while the vacuum energy is negative and independent of the gauge group.

The vacuum energy and dilaton tadpole to lowest non-trivial order for the open bosonic string. While the vacuum energy is non-zero and independent of the gauge group, the dilaton tadpole is zero for a unique choice of gauge group, SO(2^{13}) i.e. SO(8192). (From: "Dilaton Tadpole for the Open Bosonic String " Michael R. Douglas and Benjamin Grinstein - September 2,1986)

$$(1/27((1 + (19 - 2 \sqrt{29} + \sqrt{2(251 - 38 \sqrt{29})}))/ (10 e))^{15} - 233 + 4\pi + (((\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1))) - 1)^2 + 276$$

Input

$$\left(\frac{1}{27} \left(\left(1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e} \right)^{15} - \left(233 + 4\pi + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} - 1 \right)^2 + 276 \right) \right)^2$$

Exact result

$$276 + \frac{1}{729} \left(-234 + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} + \left(\frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e} \right)^{15} + 4\pi \right)^2$$

Decimal approximation

4372.0531836345975135570677492186432759533024521972168243475135145

...

4372.0531836.... $\approx$ 4372

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

From:

$$\frac{\sqrt{x^2-1} - x}{1+x} = \frac{1}{4t^2} \left\{ (1-t \cdot \frac{\sqrt{5}+1}{2}) \sqrt{1-t} - \sqrt{(1-t)(1+t \cdot \frac{\sqrt{5}+1}{2})^2 - 2t(\sqrt{5}+1)} \right. \\ \left. - (1+t \cdot \frac{\sqrt{5}-1}{2}) \sqrt{1-t} + \sqrt{(1-t)(1-t \cdot \frac{\sqrt{5}-1}{2})^2 + 2t(\sqrt{5}-1)} \right\}$$

$$\frac{1}{4t^2} \left\{ (1-t \cdot \frac{\sqrt{5}+1}{2}) \sqrt{1-t} - \sqrt{(1-t)(1+t \cdot \frac{\sqrt{5}+1}{2})^2 - 2t(\sqrt{5}+1)} \right. \\ \left. - (1+t \cdot \frac{\sqrt{5}-1}{2}) \sqrt{1-t} + \sqrt{(1-t)(1-t \cdot \frac{\sqrt{5}-1}{2})^2 + 2t(\sqrt{5}-1)} \right\}$$

$$\frac{1}{4t^2} [(1-t \cdot ((\sqrt{5}+1)/2)) \sqrt{1-t} - \sqrt{((1-t)(1+t \cdot ((\sqrt{5}+1)/2))^2 - 2t(\sqrt{5}+1))}] \\ - [(1+t \cdot ((\sqrt{5}-1)/2)) \sqrt{1-t} + \sqrt{((1-t)(1-t \cdot ((\sqrt{5}-1)/2))^2 + 2t(\sqrt{5}-1))}]$$

i.e., for $t = 0.5$:

$$\frac{1}{4 \times 25} [(1-5 \cdot ((\sqrt{5}+1)/2)) \sqrt{1-5} - \sqrt{((1-5)(1+5 \cdot ((\sqrt{5}+1)/2))^2 - 10(\sqrt{5}+1))}] \\ - [(1+5 \cdot ((\sqrt{5}-1)/2)) \sqrt{1-5} + \sqrt{((1-5)(1-5 \cdot ((\sqrt{5}-1)/2))^2 + 10(\sqrt{5}-1))}]$$

Input

$$\frac{1}{4 \times 25} \left(\left(1 - 5 \left(\frac{1}{2} (\sqrt{5} + 1) \right) \right) \sqrt{1-5} - \sqrt{(1-5) \left(1 + 5 \left(\frac{1}{2} (\sqrt{5} + 1) \right) \right)^2 - 10(\sqrt{5} + 1)} \right) \\ - \left(\left(1 + 5 \left(\frac{1}{2} (\sqrt{5} - 1) \right) \right) \sqrt{1-5} + \sqrt{(1-5) \left(1 - 5 \left(\frac{1}{2} (\sqrt{5} - 1) \right) \right)^2 + 10(\sqrt{5} - 1)} \right)$$

Result

$$\frac{1}{100} \left(\sqrt{10(\sqrt{5}-1) - 4 \left(1 - \frac{5}{2}(\sqrt{5}-1)\right)^2} - 2i \left(1 + \frac{5}{2}(\sqrt{5}-1)\right) \right) \\ \left(-\sqrt{-10(1+\sqrt{5}) - 4 \left(1 + \frac{5}{2}(1+\sqrt{5})\right)^2} + 2i \left(1 - \frac{5}{2}(1+\sqrt{5})\right) \right)$$

Decimal approximation

-1.966810914982830324127051836886016763529808285706013156181381135

...

-1.966810914982....

Alternate forms

$$\frac{1}{100} \left(2\sqrt{2(23-10\sqrt{5})} - 5\sqrt{5} + 3 \right) \left(2\sqrt{2(23+10\sqrt{5})} + 5\sqrt{5} + 3 \right)$$

$$\frac{1}{25} \left(-29 + 2\sqrt{29} - \sqrt{2(791-58\sqrt{29})} \right)$$

root of $25x^4 + 116x^3 + 66x^2 - 116x + 25$ near $x = -1.96681$
--

Minimal polynomial

$$25x^4 + 116x^3 + 66x^2 - 116x + 25$$

Expanded form

$$\begin{aligned}
& -\frac{29}{25} - \frac{1}{4}i \sqrt{\frac{1}{5} \left(10(\sqrt{5} - 1) - 4 \left(1 - \frac{5}{2}(\sqrt{5} - 1) \right)^2 \right)} - \\
& \frac{3}{100}i \sqrt{10(\sqrt{5} - 1) - 4 \left(1 - \frac{5}{2}(\sqrt{5} - 1) \right)^2} + \\
& \frac{1}{4}i \sqrt{\frac{1}{5} \left(-10(1 + \sqrt{5}) - 4 \left(1 + \frac{5}{2}(1 + \sqrt{5}) \right)^2 \right)} - \\
& \frac{3}{100}i \sqrt{-10(1 + \sqrt{5}) - 4 \left(1 + \frac{5}{2}(1 + \sqrt{5}) \right)^2} - \frac{1}{100} \\
& \sqrt{10(\sqrt{5} - 1) - 4 \left(1 - \frac{5}{2}(\sqrt{5} - 1) \right)^2} \sqrt{-10(1 + \sqrt{5}) - 4 \left(1 + \frac{5}{2}(1 + \sqrt{5}) \right)^2}
\end{aligned}$$

Multiplying the two expressions

$$\frac{1}{5} \left(-19 + 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})} \right)$$

$$\frac{1}{25} \left(-29 + 2\sqrt{29} - \sqrt{2(791 - 58\sqrt{29})} \right)$$

we obtain, after some calculations:

$$-(2+0.937)*([1/25 (-29 + 2 \sqrt{29}) - \sqrt{2 (791 - 58 \sqrt{29})}))*[1/5 (-19 + 2 \sqrt{29}) + \sqrt{2 (251 - 38 \sqrt{29})}]]$$

Input

$$\begin{aligned}
& -(2 + 0.937) \left(\left(\frac{1}{25} \left(-29 + 2\sqrt{29} - \sqrt{2(791 - 58\sqrt{29})} \right) \right) \right. \\
& \quad \left. \left(\frac{1}{5} \left(-19 + 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})} \right) \right) \right)
\end{aligned}$$

Result

1.6172420385669442657177861855964485908246265196048874648136436917

...

1.6172420385.... result that is a very good approximation to the value of the golden ratio 1.618033988749...

$$(-3*([1/25 (-29 + 2 \sqrt{29}) - \sqrt{2 (791 - 58 \sqrt{29})}]))*[1/5 (-19 + 2 \sqrt{29}) + \sqrt{2 (251 - 38 \sqrt{29})}]]^15-144+12-2((((\sqrt{10-2\sqrt{5}} -2)/(\sqrt{5}-1))))$$

Input

$$\left(-3\left(\left(\frac{1}{25}\left(-29+2\sqrt{29}-\sqrt{2\left(791-58\sqrt{29}\right)}\right)\right)\left(\frac{1}{5}\left(-19+2\sqrt{29}+\sqrt{2\left(251-38\sqrt{29}\right)}\right)\right)\right)\right)^{15}-144+12-2\times\frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}$$

Exact result

$$-132-\frac{2\left(\sqrt{10-2\sqrt{5}}-2\right)}{\sqrt{5}-1}-\frac{\left(14348907\left(-29+2\sqrt{29}-\sqrt{2\left(791-58\sqrt{29}\right)}\right)^{15}\left(-19+2\sqrt{29}+\sqrt{2\left(251-38\sqrt{29}\right)}\right)^{15}\right)}{28421709430404007434844970703125}$$

Decimal approximation

1729.0257571180384707747021060279038208699072504805436219290006352

...

1729.025757118....

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

$$(1/27((-3*([1/25 (-29 + 2 \sqrt{29}) - \sqrt{2 (791 - 58 \sqrt{29})}]))*[1/5 (-19 + 2 \sqrt{29}) + \sqrt{2 (251 - 38 \sqrt{29})}]))^{15} - 144 + 12 - 2 \left(\left(\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} - 1 \right)^2 + 276 \right)$$

Input

$$\left(\frac{1}{27} \left(\left(-3 \left(\left(\frac{1}{25} \left(-29 + 2 \sqrt{29} - \sqrt{2 (791 - 58 \sqrt{29})} \right) \right) \right. \right. \right. \right. \right. \left. \left. \left(\frac{1}{5} \left(-19 + 2 \sqrt{29} + \sqrt{2 (251 - 38 \sqrt{29})} \right) \right) \right) \right) \right)^{15} - 144 + 12 - 2 \left(\left(\frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} - 1 \right)^2 + 276 \right)$$

Exact result

$$276 + \frac{1}{729} \left(-133 - \frac{2(\sqrt{10 - 2\sqrt{5}} - 2)}{\sqrt{5} - 1} - \left(14348907 \left(-29 + 2\sqrt{29} - \sqrt{2(791 - 58\sqrt{29})} \right)^{15} \left(-19 + 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})} \right)^{15} \right) / 284217094304007434844970703125 \right)^2$$

Decimal approximation

4372.1221087289027224212442345947266523547519777670855795274800205

...

4372.1221087289... $\approx$ 4372

where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

$$(1/12(8 e^{\pi}-\pi+\ln(8)-3\ln(\pi)+6\tan^{-1}(\pi)))[((1/27((1+(19-2\sqrt{29})+\sqrt{2(251-38\sqrt{29}})))/(10e))^{15}-233+4\pi+(((\sqrt{10-2\sqrt{5}}-2)/(\sqrt{5}-1)))-1)^2+276)^2]$$

where

$$\frac{1}{12} (8 e^{\pi} - \pi + \log(8) - 3 \log(\pi) + 6 \tan^{-1}(\pi)) \approx 15.6837469855$$

Input

$$\left(\frac{1}{12} (8 e^{\pi} - \pi + \log(8) - 3 \log(\pi) + 6 \tan^{-1}(\pi)) \right) \left(\left(\frac{1}{27} \left(1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e} \right)^{15} - \left(233 + 4\pi + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} - 1 \right)^2 + 276 \right)^2 \right)$$

$\log(x)$ is the natural logarithm
 $\tan^{-1}(x)$ is the inverse tangent function

Exact Result

$$\frac{1}{12} \left(276 + \frac{1}{729} \left(-234 + \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} + \left(1 + \frac{19 - 2\sqrt{29} + \sqrt{2(251 - 38\sqrt{29})}}{10e} \right)^{15} + 4\pi \right)^{2/2} \right)$$

$$(8e^\pi - \pi + \log(8) - 3\log(\pi) + 6\tan^{-1}(\pi))$$

(result in radians)

Decimal approximation

2.99792456019043967898304295394196727278886531154016203598674... × 10⁸

(result in radians)

2.99792456019....*10⁸ ≈ c = 299792458 (speed of light)

From:

$$\mu = \frac{1}{4t} \left\{ \sqrt{1 - t^6} - \sqrt{1 - t^6 \left(\frac{\sqrt{5} + 1}{2} \right)^6} \right\} \left\{ \sqrt{1 - t^6 \left(\frac{\sqrt{5} - 1}{2} \right)^6} - \sqrt{1 - t^6} \right\}$$

$$v = \frac{1 - \mu}{1 + \mu} = \frac{1}{2t} \left\{ \sqrt{\left(\frac{\sqrt{5} + 1}{2} \right)^6 - t^6} - \sqrt{1 - t^6} \right\} \left\{ \sqrt{\left(\frac{\sqrt{5} - 1}{2} \right)^6 - t^6} + \sqrt{1 - t^6} \right\}$$

For t = 2:

$$\mu = \frac{1}{4t} \left\{ \sqrt{1 - t^6} - \sqrt{1 - t^6 \left(\frac{\sqrt{5} + 1}{2} \right)^6} \right\} \left\{ \sqrt{1 - t^6 \left(\frac{\sqrt{5} - 1}{2} \right)^6} - \sqrt{1 - t^6} \right\}$$

From:

$$\frac{1}{4t} \left\{ \sqrt{1 - t^6} - \sqrt{1 - t^6 \left(\frac{\sqrt{5} + 1}{2} \right)^6} \right\} \left\{ \sqrt{1 - t^6 \left(\frac{\sqrt{5} - 1}{2} \right)^6} - \sqrt{1 - t^6} \right\}$$

$$\frac{1}{4 \times 2^6} [\sqrt{1-2^6} - \sqrt{1-2^6 \left(\frac{\sqrt{5}+1}{2} \right)^6}] [\sqrt{1-2^6 \left(\frac{\sqrt{5}-1}{2} \right)^6} - \sqrt{1-2^6}]$$

Input

$$\frac{1}{4 \times 2^6} \left(\sqrt{1-2^6} - \sqrt{1-2^6 \left(\frac{1}{2} (\sqrt{5}+1) \right)^6} \right) \left(\sqrt{1-2^6 \left(\frac{1}{2} (\sqrt{5}-1) \right)^6} - \sqrt{1-2^6} \right)$$

Result

$$\frac{1}{256} \left(i \sqrt{(\sqrt{5}-1)^6 - 1} - 3i\sqrt{7} \right) \left(3i\sqrt{7} - i \sqrt{(1+\sqrt{5})^6 - 1} \right)$$

Decimal approximation

$$-0.641847463248368221439538308482465306879031653471522075442432698$$

...

$$-0.64184746324....$$

Alternate forms

$$\frac{1}{256} \left(\sqrt{575 + 256\sqrt{5}} - 3\sqrt{7} \right) \left(\sqrt{575 - 256\sqrt{5}} - 3\sqrt{7} \right)$$

$$\frac{1}{256} \left(63 + \sqrt{2945} - 3\sqrt{14(575 + \sqrt{2945})} \right)$$

$\text{root of } 64x^4 - 63x^3 - 124x^2 + 63x + 64 \text{ near } x = -0.641847$

Minimal polynomial

$$64x^4 - 63x^3 - 124x^2 + 63x + 64$$

Expanded forms

$$-\frac{3}{256} \sqrt{7(575 - 256\sqrt{5})} - \frac{3}{256} \sqrt{7(575 + 256\sqrt{5})} + \frac{1}{256} \left(63 + \sqrt{(575 - 256\sqrt{5})(575 + 256\sqrt{5})} \right)$$

$$\frac{63}{256} - \frac{3}{256} \sqrt{7((\sqrt{5} - 1)^6 - 1)} - \frac{3}{256} \sqrt{7((1 + \sqrt{5})^6 - 1)} + \frac{1}{256} \sqrt{((\sqrt{5} - 1)^6 - 1)((1 + \sqrt{5})^6 - 1)}$$

From which:

$$1 - \frac{1}{4 \times 2^6} [\sqrt{1 - 2^6} - \sqrt{1 - 2^6 \left(\frac{\sqrt{5} + 1}{2} \right)^6}] [\sqrt{1 - 2^6 \left(\frac{\sqrt{5} - 1}{2} \right)^6} - \sqrt{1 - 2^6}]$$

Input

$$1 - \frac{1}{4 \times 2^6} \left(\sqrt{1 - 2^6} - \sqrt{1 - 2^6 \left(\frac{1}{2} (\sqrt{5} + 1) \right)^6} \right) \left(\sqrt{1 - 2^6 \left(\frac{1}{2} (\sqrt{5} - 1) \right)^6} - \sqrt{1 - 2^6} \right)$$

Result

$$1 - \frac{1}{256} \left(i \sqrt{(\sqrt{5} - 1)^6 - 1} - 3i \sqrt{7} \right) \left(3i \sqrt{7} - i \sqrt{(1 + \sqrt{5})^6 - 1} \right)$$

Decimal approximation

1.6418474632483682214395383084824653068790316534715220754424326983

...

$$1.6418474632\dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934\dots$$

Alternate forms

$$\frac{1}{256} \left(3 \sqrt{7(575 - 256\sqrt{5})} + 3 \sqrt{7(575 + 256\sqrt{5})} - \sqrt{(575 - 256\sqrt{5})(575 + 256\sqrt{5}) + 193} \right)$$

$$\frac{1}{256} \left(193 - \sqrt{2945} + 3 \sqrt{14(575 + \sqrt{2945})} \right)$$

root of $64x^4 - 193x^3 + 71x^2 + 118x + 4$ near $x = 1.64185$

Minimal polynomial

$$64x^4 - 193x^3 + 71x^2 + 118x + 4$$

Expanded forms

$$\frac{193}{256} + \frac{3}{256} \sqrt{7(575 - 256\sqrt{5})} + \frac{3}{256} \sqrt{7(575 + 256\sqrt{5})} - \frac{1}{256} \sqrt{(575 - 256\sqrt{5})(575 + 256\sqrt{5})}$$

$$\frac{193}{256} + \frac{3}{256} \sqrt{7((\sqrt{5}-1)^6-1)} + \frac{3}{256} \sqrt{7((1+\sqrt{5})^6-1)} - \frac{1}{256} \sqrt{((\sqrt{5}-1)^6-1)((1+\sqrt{5})^6-1)}$$

From:

$$\frac{1}{4} \left\{ \sqrt{\left(\frac{\sqrt{5}+1}{2}\right)^6 - 2^6} - \sqrt{1-2^6} \right\} \left\{ \sqrt{\left(\frac{\sqrt{5}-1}{2}\right)^6 - 2^6} + \sqrt{1-2^6} \right\}$$

$$\frac{1}{4} [\text{sqrt}(((\text{sqrt}5+1)/2)^6-2^6) - \text{sqrt}(1-2^6)] [\text{sqrt}(((\text{sqrt}5-1)/2)^6-2^6) + \text{sqrt}(1-2^6)]$$

Input

$$\frac{1}{4} \left(63 - \sqrt{2945} + 3 \sqrt{14(55 - \sqrt{2945})} \right)$$

Decimal approximation

4.5842128556161561270448016647608722854929374633141487839024408609

...

4.584212855....

Alternate forms

$$\frac{3}{4} \sqrt{770 - 14 \sqrt{2945}} - \sqrt{2945} \frac{1}{4} + \frac{63}{4}$$

$$\frac{1}{4} \left(63 - \sqrt{2945} + 3 \sqrt{770 - 14 \sqrt{2945}} \right)$$

root of $x^4 - 63x^3 + 254x^2 + 63x + 1$ near $x = 4.58421$

Minimal polynomial

$$x^4 - 63x^3 + 254x^2 + 63x + 1$$

Expanded form

$$\frac{63}{4} - \frac{\sqrt{2945}}{4} + \frac{3}{2} \sqrt{\frac{7}{2} (55 - \sqrt{2945})}$$

$$(((1/256 (63 + \sqrt{2945}) - 3 \sqrt{14 (575 + \sqrt{2945})}))) * ((1/4 (63 - \sqrt{2945}) + 3 \sqrt{14 (55 - \sqrt{2945})}))))$$

Input

$$-\left(\left(\frac{1}{256} \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})}\right)\right)\left(\frac{1}{4} \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})}\right)\right)\right)$$

Result

$$-\frac{\left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})}\right) \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})}\right)}{1024}$$

Decimal approximation

2.9423653923677879056052633562784069786139058098426267084600081626

...

2.942365392....

Alternate forms

$$\frac{1}{1024} \left(-189 \sqrt{14 (55 - \sqrt{2945})} - 252 \sqrt{2945} - 3 \sqrt{41230 (55 - \sqrt{2945})} + \right. \\ \left. 189 \sqrt{14 (575 + \sqrt{2945})} - 3 \sqrt{41230 (575 + \sqrt{2945})} + 15356 \right)$$

$$\frac{1}{256} \left(3839 - 63 \sqrt{2945} + 3 \sqrt{14 (209215 - 3839 \sqrt{2945})} \right)$$

root of $64x^4 - 3839x^3 + 12039x^2 - 3839x + 64$ near $x = 2.94237$
--

Minimal polynomial

$$64x^4 - 3839x^3 + 12039x^2 - 3839x + 64$$

Expanded forms

$$\frac{3839}{256} - \frac{63 \sqrt{2945}}{256} - \frac{3 \sqrt{2267650 - 41230 \sqrt{2945}}}{1024} - \frac{189 \sqrt{770 - 14 \sqrt{2945}}}{1024} + \\ \frac{189 \sqrt{8050 + 14 \sqrt{2945}}}{1024} - \frac{3 \sqrt{23707250 + 41230 \sqrt{2945}}}{1024}$$

$$-1 - \frac{189}{512} \sqrt{\frac{7}{2} (55 - \sqrt{2945})} - \\ \frac{3}{512} \sqrt{\frac{20615}{2} (55 - \sqrt{2945})} + \frac{189}{512} \sqrt{\frac{7}{2} (575 + \sqrt{2945})} - \\ \frac{3}{512} \sqrt{\frac{20615}{2} (575 + \sqrt{2945})} + \frac{63}{512} \sqrt{(55 - \sqrt{2945}) (575 + \sqrt{2945})}$$

From

$$- \frac{\left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})}\right) \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})}\right)}{1024}$$

we obtain:

$$4((((-(63 - \sqrt{2945}) + 3 \sqrt{14 (55 - \sqrt{2945})})) (63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})})) / 2.94236539236)))$$

Input interpretation

$$4 \left(- \frac{1}{2.94236539236} \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right)$$

Result

4096.00000001...

4096.00000001.... $\approx 4096 = 64^2$

$$4((((-(63 - \sqrt{2945}) + 3 \sqrt{14 (55 - \sqrt{2945})})) (63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})})) / 2.94236539236))) + 276$$

Input interpretation

$$4 \left(- \frac{1}{2.94236539236} \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) + 276$$

Result

4372.00000001...

4372.00000001.... where 4372 is a value indicated in the fundamental Ramanujan paper “**Modular equations and Approximations to π** ”

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

27sqrt(((4(((63 - sqrt(2945) + 3 sqrt(14 (55 - sqrt(2945)))) (63 + sqrt(2945) - 3
sqrt(14 (575 + sqrt(2945)))))) / 2.94236539236)))))+1

Input interpretation

$$27 \sqrt{\left(4 \left(-\frac{1}{2.94236539236} \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})}\right) \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})}\right)\right)\right) + 1}$$

Result

1729.000000000...

1729

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

$$(((27\sqrt{4\left(-\frac{1}{2.94236539236}\left(63 - \sqrt{2945} + 3\sqrt{14(55 - \sqrt{2945})}\right)}\right)}\left(63 + \sqrt{2945} - 3\sqrt{14(575 + \sqrt{2945})}\right)} + 1)))^{1/15}$$

Input interpretation

$$\left(27\sqrt{4\left(-\frac{1}{2.94236539236}\left(63 - \sqrt{2945} + 3\sqrt{14(55 - \sqrt{2945})}\right)}\right)}\left(63 + \sqrt{2945} - 3\sqrt{14(575 + \sqrt{2945})}\right)} + 1\right)^{(1/15)}$$

Result

1.643815228749...

$$1.643815228749\dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$$

$$\left(\frac{1}{48} e^8 \left(\frac{1}{6} \log(2) \log(3)\right)^{2/3}\right) \left(\left(\left(4\left(-\frac{1}{2.94236539236}\left(63 - \sqrt{2945} + 3\sqrt{14(55 - \sqrt{2945})}\right)}\right)}\left(63 + \sqrt{2945} - 3\sqrt{14(575 + \sqrt{2945})}\right)} + 276\right)\right)^2$$

where

$$\frac{1}{48} e^8 \left(\frac{1}{6} \log(2) \log(3)\right)^{2/3} \approx 15.68412855495$$

Input interpretation

$$\left(\frac{1}{48} e^8 \left(\frac{1}{6} \log(2) \log(3)\right)^{2/3}\right) \left(4\left(-\frac{1}{2.94236539236}\left(63 - \sqrt{2945} + 3\sqrt{14(55 - \sqrt{2945})}\right)}\right)\left(63 + \sqrt{2945} - 3\sqrt{14(575 + \sqrt{2945})}\right)} + 276\right)^2$$

$\log(x)$ is the natural logarithm

Result

$$2.99792455906... \times 10^8$$

$$2.99792455906... * 10^8 \approx c$$

Alternative representations

$$\begin{aligned} & \frac{1}{48} \left(\left(4 \left(- \left(\left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \right. \right. \right. \right. \\ & \quad \left. \left. \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) \right) \right) / \\ & \quad \left. 2.942365392360000 + 276 \right)^2 \left(e^8 \left(\frac{1}{6} \log(2) \log(3) \right)^{2/3} \right) = \\ & \frac{1}{48} e^8 \left(\frac{\log_e(2) \log_e(3)}{6} \right)^{2/3} \left(276 - \left(4 \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \left(63 + \right. \right. \right. \\ & \quad \left. \left. \left. \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) \right) / 2.942365392360000 \right)^2 \end{aligned}$$

$$\begin{aligned} & \frac{1}{48} \left(\left(4 \left(- \left(\left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \right. \right. \right. \right. \\ & \quad \left. \left. \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) \right) \right) / \\ & \quad \left. 2.942365392360000 + 276 \right)^2 \left(e^8 \left(\frac{1}{6} \log(2) \log(3) \right)^{2/3} \right) = \\ & \frac{1}{48} e^8 \left(\frac{1}{6} \log^2(a) \log_a(2) \log_a(3) \right)^{2/3} \\ & \quad \left(276 - \left(4 \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \left(63 + \sqrt{2945} - \right. \right. \right. \\ & \quad \left. \left. \left. 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) \right) / 2.942365392360000 \right)^2 \end{aligned}$$

$$\begin{aligned}
& \frac{1}{48} \left(\left(4 \left(- \left(\left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \right. \right. \right. \right. \\
& \quad \left. \left. \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) \right) \right) / \\
& \quad \left. 2.942365392360000 + 276 \right)^2 \left(e^8 \left(\frac{1}{6} \log(2) \log(3) \right)^{2/3} \right) = \\
& \frac{1}{48} e^8 \left(\frac{4}{6} \coth^{-1}(2) \coth^{-1}(3) \right)^{2/3} \\
& \quad \left(276 - \left(4 \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \left(63 + \sqrt{2945} - \right. \right. \right. \\
& \quad \left. \left. \left. 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) / 2.942365392360000 \right)^2
\end{aligned}$$

Integral representations

$$\begin{aligned}
& \frac{1}{48} \left(\left(4 \left(- \left(\left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \right. \right. \right. \right. \\
& \quad \left. \left. \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) \right) \right) / \\
& \quad \left. 2.942365392360000 + 276 \right)^2 \left(e^8 \left(\frac{1}{6} \log(2) \log(3) \right)^{2/3} \right) = \\
& \frac{1}{48 \times 6^{2/3}} e^8 \left(\int_0^1 \int_0^1 \frac{1}{(1+t_1)(1+2t_2)} dt_2 dt_1 \right)^{2/3} \\
& \quad \left(276 + 1.359450464713255 \left(-63.000000000000000 + \sqrt{2945} - \right. \right. \\
& \quad \left. \left. 3.000000000000000 \sqrt{770 - 14 \sqrt{2945}} \right) \left(63.000000000000000 + \right. \right. \\
& \quad \left. \left. \sqrt{2945} - 3.000000000000000 \sqrt{14 (575 + \sqrt{2945})} \right) \right)^2
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{48} \left(\left(4 \left(- \left(\left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \right. \right. \right. \right. \\
& \quad \left. \left. \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right) \right) \right) / \\
& \quad \left. 2.942365392360000 + 276 \right)^2 \left(e^8 \left(\frac{1}{6} \log(2) \log(3) \right)^{2/3} \right) = \\
& \frac{1}{192 \times 3^{2/3}} e^8 \left(\frac{\left(\int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{\Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds \right) \left(\int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{2^{-s} \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds \right)^{2/3}}{i^2 \pi^2} \right) \\
& \quad \left(276 - 1.359450464713255 \left(63 - \sqrt{2945} + 3 \sqrt{14 (55 - \sqrt{2945})} \right) \right. \\
& \quad \left. \left(63 + \sqrt{2945} - 3 \sqrt{14 (575 + \sqrt{2945})} \right) \right)^2 \text{ for } -1 < \gamma < 0
\end{aligned}$$

We know that the one-loop quantum-mechanical fluctuations about the instanton have been explicitly calculated by 't Hooft [1977], who showed that the instanton solved the long-standing $U(1)$ problem via its coupling to the anomaly of the ninth axial current.

Furthermore, in oriented string theories, a Dp brane is a gauge theory instanton in the world volume $(p + 5)$ -dimensional $U(N)$ gauge theory on a stack of N D($p + 4$)-branes.

From Wikipedia:

Note that there is sometimes also a corresponding soliton in a theory with one additional space dimension. Recent research on *instantons* links them to topics such as D-branes and Black holes and, of course, the vacuum structure of QCD. For example, in oriented string theories, a Dp brane is a gauge theory instanton in the world volume $(p + 5)$ -dimensional $U(N)$ gauge theory on a stack of N D($p + 4$)-branes.

From:

Instantons and Solitons - <https://eduardo.physics.illinois.edu/phys583/ch19.pdf>

We have that:

On the other hand, we saw that in the abelian Higgs model, a theory of a complex scalar field coupled to the a $U(1)$ gauge field, has vortex classical regular vortex solutions in the classically spontaneously broken phase of this theory. G. 't Hooft ('t Hooft, 1976) and A. M. Polyakov (Polyakov, 1975a) showed that a regular configuration which at long distances becomes a Dirac magnetic monopole exists in the Higgs sector of the Georgi-Glashow model (Georgi and Glashow, 1974), that we already discussed in section 18.11.2.

The Georgi-Glashow model has a three-component real scalar field $\phi = (\phi_1, \phi_2, \phi_3)$, that transforms under the adjoint representation of $SU(2)$, and a Yang-Mills gauge field taking values in the algebra of the gauge subgroup $G = SU(2)$ associated with the weak interactions of a grand unified gauge theory with gauge group $SU(5)$. In the spontaneously broken phase, the Lagrangian in $D = 2 + 1$ -dimensional Euclidean space is

$$\mathcal{L} = \frac{1}{2}(D_\mu \phi)^2 - \frac{m^2}{2}\phi^2 + \frac{\lambda}{4!}(\phi^2)^2 + \frac{1}{4}\text{tr}F_{\mu\nu}^2 \quad (19.154)$$

where $D_\mu \phi = \partial_\mu \phi + gA_\mu \times \phi$ is the covariant derivative in the adjoint representation of $SU(2)$. In this phase, the $SU(2)$ gauge symmetry is spontaneously broken down to its $U(1)$ subgroup. As a subgroup of the compact Lie group $SU(2)$, the unbroken $U(1)$ subgroup is compact.

The classical (Euclidean) equations of motion are (with $a = 1, 2, 3$ and $i, j = 1, 2, 3$)

$$D_i F^{aij} = g\epsilon^{abc}(D_j \phi_b)\phi_c \quad (19.155)$$

and

$$D_i D_i \phi_a = -\lambda \phi^2 \phi_a + \lambda f^2 \phi_a \quad (19.156)$$

where $\lambda f^2 = |m^2|$. The Euclidean action of the $D = 3$ dimensional instanton (or the energy of the $D = 3$ soliton) is

$$S_E = \int d^3x \left[\frac{1}{4}F_{ij}^a F^{aij} + \frac{1}{2}D_i \phi_a D_i \phi_a + \frac{\lambda}{4!}(\phi^2 - f^2)^2 \right] \quad (19.157)$$

Let us first look at the zero energy solutions, the classical vacuum. They are $A_i^a = 0$, $\phi^2 = f^2$, and $D_i \phi = 0$. In this case, the latter reduces to $\partial_i \phi = 0$ and ϕ is a constant vector.

From:

Computation of the quantum effects due to a four-dimensional pseudo-particle.

– Gerard 't Hooft - Phys. Rev. **D14** (1976) 3432, reprinted in: *Series of Selected Papers in Physics 70*, Physical Society of Japan, THEORY OF GAUGE FIELDS, (K. Kikkawa and A. Hosoya eds.), pp 217-235.

We have that :

$$\begin{aligned} R &= \frac{1}{12} - \zeta'(-1) \\ &= \frac{\ln 2\pi + \gamma}{12} - \frac{\zeta'(2)}{2\pi^2} \\ &= \frac{1}{12} (\ln 2\pi + \gamma) + \frac{1}{2\pi^2} \sum_{s=1}^{\infty} \frac{\ln s}{s^2} , \end{aligned}$$

$\gamma = 0.577\,215\,664\,9$ is Euler's constant,

and

$$\begin{aligned} -\zeta'(2) &= \sum \frac{\ln s}{s^2} \\ &= 0.937\,548\,254\,315\,844 . \end{aligned}$$

TABLE I. Numerical values of the coefficients $C(t)$ and $\alpha(t)$ as they occur in the text.

t	$C(t)$	$\alpha(t)$
0	0	0
$\frac{1}{2}$	1	$2R - \frac{1}{6} \ln 2 - \frac{17}{72} = 0.145\,873$
1	4	$8R + \frac{1}{3} \ln 2 - \frac{16}{9} = 0.443\,307$
$\frac{3}{2}$	10	$20R + 4 \ln 3 - \frac{5}{3} \ln 2 - \frac{265}{36} = 0.853\,182$
$R = \frac{1}{12} (\ln 2\pi + \gamma) + \frac{1}{2\pi^2} \sum_2^{\infty} \frac{\ln s}{s^2} = 0.248\,754\,477\,033\,784$		

We have:

$$2^{14} \pi^6 g^{-8} \rho^{-5} d^4 z d\rho \exp \left\{ -\frac{8\pi^2}{g_B^2(n)} + \left(\ln \rho + \frac{1}{4-n} \right) \left[\frac{22}{3} - \frac{1}{6} \sum_t N^s(t) C(t) \right] + A - \sum_t N^s(t) A(t) \right\}, \quad (13.8)$$

And:

$$\begin{aligned} A &= -\alpha(1) + \frac{11}{3} (\ln 4\pi - \gamma) + \frac{5}{18} = 6.998\,435 \\ A(t) &= \alpha(t) + \frac{1}{12} (\ln 4\pi - \gamma - \frac{5}{6}) C(t). \\ A(0) &= 0, \quad A(\frac{1}{2}) = 0.239\,246, \quad A(1) = 0.816\,799, \\ A(\frac{3}{2}) &= 1.786\,912. \end{aligned} \quad (13.10)$$

$$-(8 \times 0.248754477033784 + \frac{1}{3} \ln(2) - \frac{16}{9}) + \frac{11}{3} (\ln(4\pi) - 0.5772156649) + \frac{5}{18}$$

Input interpretation

$$-\left(8 \times 0.248754477033784 + \frac{1}{3} \log(2) - \frac{16}{9}\right) + \frac{11}{3} (\log(4\pi) - 0.5772156649) + \frac{5}{18}$$

$\log(x)$ is the natural logarithm

Result

6.9984354800...

6.9984354800...

Alternative representations

$$\begin{aligned} & -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = \\ & -1.990035816270272000 - \frac{\log_e(2)}{3} + \frac{11}{3} (-0.577216 + \log_e(4 \pi)) + \frac{16}{9} + \frac{5}{18} \end{aligned}$$

$$\begin{aligned} & -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = \\ & -1.990035816270272000 - \frac{1}{3} \log(a) \log_a(2) + \\ & \frac{11}{3} (-0.577216 + \log(a) \log_a(4 \pi)) + \frac{16}{9} + \frac{5}{18} \end{aligned}$$

$$\begin{aligned} & -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = \\ & -1.990035816270272000 + \frac{\text{Li}_1(-1)}{3} + \frac{11}{3} (-0.577216 - \text{Li}_1(1 - 4 \pi)) + \frac{16}{9} + \frac{5}{18} \end{aligned}$$

Series representations

$$\begin{aligned} & -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = \\ & -2.05094 - 0.666667 i \pi \left\lfloor \frac{\arg(2 - x)}{2 \pi} \right\rfloor + 7.33333 i \pi \left\lfloor \frac{\arg(4 \pi - x)}{2 \pi} \right\rfloor + 3.33333 \\ & \log(x) + \sum_{k=1}^{\infty} \frac{(-1)^k (0.333333 (2 - x)^k - 3.66667 (4 \pi - x)^k) x^{-k}}{k} \quad \text{for } x < 0 \end{aligned}$$

$$\begin{aligned}
& -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = \\
& -2.05094 - 0.666667 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + \\
& 7.33333 i \pi \left[-\frac{-\pi + \arg\left(\frac{4 \pi}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + 3.33333 \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k (0.333333 (2 - z_0)^k - 3.66667 (4 \pi - z_0)^k) z_0^{-k}}{k}
\end{aligned}$$

$$\begin{aligned}
& -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = \\
& -2.05094 - \frac{1}{3} \left[\frac{\arg(2 - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \\
& \frac{11}{3} \left[\frac{\arg(4 \pi - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \frac{10 \log(z_0)}{3} - \frac{1}{3} \left[\frac{\arg(2 - z_0)}{2 \pi} \right] \log(z_0) + \\
& \frac{11}{3} \left[\frac{\arg(4 \pi - z_0)}{2 \pi} \right] \log(z_0) + \sum_{k=1}^{\infty} \frac{(-1)^k ((2 - z_0)^k - 11 (4 \pi - z_0)^k) z_0^{-k}}{3 k}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = \\
& -2.05094 + \int_1^2 \left(\frac{3.66667 (-1 + 4 \pi)}{2 + 4 \pi (-1 + t) - t} - \frac{0.333333}{t} \right) dt
\end{aligned}$$

$$\begin{aligned}
& -\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \\
& \frac{1}{3} (\log(4 \pi) - 0.577216) 11 + \frac{5}{18} = -2.05094 + \\
& \int_{-i \infty + \gamma}^{i \infty + \gamma} -\frac{0.166667 (-1 + 4 \pi)^{-s} (-11 + (-1 + 4 \pi)^s) \Gamma(-s)^2 \Gamma(1 + s)}{i \pi \Gamma(1 - s)} ds \\
& \text{for } -1 < \gamma < 0
\end{aligned}$$

$$(2 \times 0.248754477033784 - \frac{1}{6} \ln(2) - \frac{17}{72}) + \frac{1}{12}(\ln(4\pi) - 0.5772156649 - \frac{5}{6})$$

Input interpretation

$$2 \times 0.248754477033784 - \frac{1}{6} \log(2) - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.5772156649 - \frac{5}{6} \right)$$

$\log(x)$ is the natural logarithm

Result

0.23924625026...

0.23924625026

Alternative representations

$$\begin{aligned} 2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = \\ 0.4975089540675680000 - \frac{\log_e(2)}{6} + \frac{1}{12} \left(-0.577216 + \log_e(4\pi) - \frac{5}{6} \right) - \frac{17}{72} \end{aligned}$$

$$\begin{aligned} 2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = \\ 0.4975089540675680000 - \frac{1}{6} \log(a) \log_a(2) + \\ \frac{1}{12} \left(-0.577216 + \log(a) \log_a(4\pi) - \frac{5}{6} \right) - \frac{17}{72} \end{aligned}$$

$$\begin{aligned} 2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = \\ 0.4975089540675680000 + \frac{\text{Li}_1(-1)}{6} + \frac{1}{12} \left(-0.577216 - \text{Li}_1(1 - 4\pi) - \frac{5}{6} \right) - \frac{17}{72} \end{aligned}$$

Series representations

$$\begin{aligned}
& 2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = \\
& 0.143852 - 0.333333 i \pi \left[\frac{\arg(2-x)}{2\pi} \right] + \\
& 0.166667 i \pi \left[\frac{\arg(4\pi-x)}{2\pi} \right] - 0.0833333 \log(x) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k (0.166667 (2-x)^k - 0.0833333 (4\pi-x)^k) x^{-k}}{k} \quad \text{for } x < 0
\end{aligned}$$

$$\begin{aligned}
& 2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = \\
& 0.143852 - 0.333333 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2\pi} \right] + \\
& 0.166667 i \pi \left[-\frac{-\pi + \arg\left(\frac{4\pi}{z_0}\right) + \arg(z_0)}{2\pi} \right] - 0.0833333 \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k (0.166667 (2-z_0)^k - 0.0833333 (4\pi-z_0)^k) z_0^{-k}}{k}
\end{aligned}$$

$$\begin{aligned}
& 2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = \\
& 0.143852 - \frac{1}{6} \left[\frac{\arg(2-z_0)}{2\pi} \right] \log\left(\frac{1}{z_0}\right) + \\
& \frac{1}{12} \left[\frac{\arg(4\pi-z_0)}{2\pi} \right] \log\left(\frac{1}{z_0}\right) - \frac{\log(z_0)}{12} - \frac{1}{6} \left[\frac{\arg(2-z_0)}{2\pi} \right] \log(z_0) + \\
& \frac{1}{12} \left[\frac{\arg(4\pi-z_0)}{2\pi} \right] \log(z_0) + \sum_{k=1}^{\infty} \frac{(-1)^k (2(2-z_0)^k - (4\pi-z_0)^k) z_0^{-k}}{12k}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& 2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = \\
& 0.143852 + \int_1^2 \left(\frac{0.0833333 (-1+4\pi)}{2+4\pi(-1+t)-t} - \frac{0.166667}{t} \right) dt
\end{aligned}$$

$$2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) = 0.143852 + \int_{-i\infty+\gamma}^{i\infty+\gamma} - \frac{0.0833333 (-1+4\pi)^{-s} (-0.5+(-1+4\pi)^s) \Gamma(-s)^2 \Gamma(1+s)}{i\pi \Gamma(1-s)} ds$$

for $-1 < \gamma < 0$

$$(8 \times 0.248754477033784 + \frac{1}{3} \ln(2) - \frac{16}{9}) + \frac{1}{12} (\ln(4\pi) - 0.5772156649 - \frac{5}{6}) \times 4$$

Input interpretation

$$\left(8 \times 0.248754477033784 + \frac{1}{3} \log(2) - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.5772156649 - \frac{5}{6} \right) \times 4$$

$\log(x)$ is the natural logarithm

Result

0.81679884826...

0.81679884826

Alternative representations

$$\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) 4 = 1.990035816270272000 + \frac{\log_e(2)}{3} - \frac{16}{9} + \frac{4}{12} \left(-0.577216 + \log_e(4\pi) - \frac{5}{6} \right)$$

$$\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) 4 = 1.990035816270272000 + \frac{1}{3} \log(a) \log_a(2) - \frac{16}{9} + \frac{4}{12} \left(-0.577216 + \log(a) \log_a(4\pi) - \frac{5}{6} \right)$$

$$\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) 4 = 1.990035816270272000 - \frac{\text{Li}_1(-1)}{3} - \frac{16}{9} + \frac{4}{12} \left(-0.577216 - \text{Li}_1(1-4\pi) - \frac{5}{6} \right)$$

Series representations

$$\begin{aligned} & \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) 4 = \\ & -0.257925 + 0.666667 i \pi \left[\frac{\arg(2-x)}{2\pi} \right] + 0.666667 i \pi \left[\frac{\arg(4\pi-x)}{2\pi} \right] + \\ & 0.666667 \log(x) + 0.333333 \sum_{k=1}^{\infty} \frac{(-1)^{1+k} ((2-x)^k + (4\pi-x)^k) x^{-k}}{k} \quad \text{for } x < 0 \end{aligned}$$

$$\begin{aligned} & \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) 4 = \\ & -0.257925 + \frac{1}{3} \left[\frac{\arg(2-z_0)}{2\pi} \right] \log\left(\frac{1}{z_0}\right) + \\ & \frac{1}{3} \left[\frac{\arg(4\pi-z_0)}{2\pi} \right] \log\left(\frac{1}{z_0}\right) + \frac{2 \log(z_0)}{3} + \frac{1}{3} \left[\frac{\arg(2-z_0)}{2\pi} \right] \log(z_0) + \\ & \frac{1}{3} \left[\frac{\arg(4\pi-z_0)}{2\pi} \right] \log(z_0) + \frac{1}{3} \sum_{k=1}^{\infty} \frac{(-1)^{1+k} ((2-z_0)^k + (4\pi-z_0)^k) z_0^{-k}}{k} \end{aligned}$$

$$\begin{aligned} & \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) 4 = \\ & -0.257925 + 0.666667 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2\pi} \right] + \\ & 0.666667 i \pi \left[-\frac{-\pi + \arg\left(\frac{4\pi}{z_0}\right) + \arg(z_0)}{2\pi} \right] + 0.666667 \log(z_0) + \\ & 0.333333 \sum_{k=1}^{\infty} \frac{(-1)^{1+k} ((2-z_0)^k + (4\pi-z_0)^k) z_0^{-k}}{k} \end{aligned}$$

Integral representations

$$\begin{aligned} & \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6} \right) 4 = \\ & -0.257925 + \int_1^2 \left(\frac{0.333333 (-1+4\pi)}{2+4\pi(-1+t)-t} + \frac{0.333333}{t} \right) dt \end{aligned}$$

$$\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) + \frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6}\right) 4 =$$

$$-0.257925 + \frac{0.166667}{i\pi} \int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{(-1+4\pi)^{-s} (1+(-1+4\pi)^s) \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds$$

for $-1 < \gamma < 0$

$$(20 \times 0.248754477033784 + 4 \ln(3) - \frac{5}{3} \ln(2) - \frac{265}{36}) + \frac{1}{12} (\ln(4\pi) - 0.5772156649 - \frac{5}{6}) \times 10$$

Input interpretation

$$\left(20 \times 0.248754477033784 + 4 \log(3) - \frac{5}{3} \log(2) - \frac{265}{36}\right) +$$

$$\frac{1}{12} \left(\log(4\pi) - 0.5772156649 - \frac{5}{6}\right) \times 10$$

$\log(x)$ is the natural logarithm

Result

1.7869116573...

1.7869116573

Alternative representations

$$\left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) +$$

$$\frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6}\right) 10 = 4.975089540675680000 - \frac{5}{3} \log(a) \log_a(2) +$$

$$4 \log(a) \log_a(3) + \frac{10}{12} \left(-0.577216 + \log(a) \log_a(4\pi) - \frac{5}{6}\right) - \frac{265}{36}$$

$$\left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) +$$

$$\frac{1}{12} \left(\log(4\pi) - 0.577216 - \frac{5}{6}\right) 10 = 4.975089540675680000 -$$

$$\frac{5 \log_e(2)}{3} + 4 \log_e(3) + \frac{10}{12} \left(-0.577216 + \log_e(4\pi) - \frac{5}{6}\right) - \frac{265}{36}$$

$$\begin{aligned} & \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\ & \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = 4.975089540675680000 - \\ & 4 \operatorname{Li}_1(-2) + \frac{5 \operatorname{Li}_1(-1)}{3} + \frac{10}{12} \left(-0.577216 - \operatorname{Li}_1(1 - 4 \pi) - \frac{5}{6} \right) - \frac{265}{36} \end{aligned}$$

Series representations

$$\begin{aligned} & \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\ & \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = \\ & -3.56148 - \frac{10}{3} i \pi \left[\frac{\arg(2-x)}{2 \pi} \right] + 8 i \pi \left[\frac{\arg(3-x)}{2 \pi} \right] + \frac{5}{3} i \pi \left[\frac{\arg(4 \pi - x)}{2 \pi} \right] + \\ & \frac{19 \log(x)}{6} + \sum_{k=1}^{\infty} \frac{(-1)^k (10(2-x)^k - 24(3-x)^k - 5(4 \pi - x)^k) x^{-k}}{6 k} \quad \text{for } x < 0 \end{aligned}$$

$$\begin{aligned} & \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\ & \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = \\ & -3.56148 - \frac{5}{3} \left[\frac{\arg(2-z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + 4 \left[\frac{\arg(3-z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \\ & \frac{5}{6} \left[\frac{\arg(4 \pi - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \frac{19 \log(z_0)}{6} - \frac{5}{3} \left[\frac{\arg(2-z_0)}{2 \pi} \right] \log(z_0) + \\ & 4 \left[\frac{\arg(3-z_0)}{2 \pi} \right] \log(z_0) + \frac{5}{6} \left[\frac{\arg(4 \pi - z_0)}{2 \pi} \right] \log(z_0) + \\ & \sum_{k=1}^{\infty} \frac{(-1)^k (10(2-z_0)^k - 24(3-z_0)^k - 5(4 \pi - z_0)^k) z_0^{-k}}{6 k} \end{aligned}$$

$$\begin{aligned}
& \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\
& \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = -3.56148 - \\
& 3.33333 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + 8 i \pi \left[-\frac{-\pi + \arg\left(\frac{3}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + \\
& 1.66667 i \pi \left[-\frac{-\pi + \arg\left(\frac{4\pi}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + 3.16667 \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k \left(1.66667 (2 - z_0)^k - 4 (3 - z_0)^k - 0.833333 (4 \pi - z_0)^k \right) z_0^{-k}}{k}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\
& \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = -3.56148 + \\
& \int_1^{4\pi} \frac{2.5 + 6.66667 \pi^2 - 8.41667 t + 3.16667 t^2 + \pi (-8.33333 + 17.6667 t)}{t (-2 + 4 \pi + t) (-1.5 + 2 \pi + t)} dt
\end{aligned}$$

$$\begin{aligned}
& \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\
& \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = \\
& -3.56148 + \int_{-i \infty + \gamma}^{i \infty + \gamma} \frac{1}{i \pi \Gamma(1 - s)} (-2 + 8 \pi)^{-s} \left(0.416667 \times 2^s + 2. (-1 + 4 \pi)^s - \right. \\
& \left. 0.833333 (-2 + 8 \pi)^s \right) \Gamma(-s)^2 \Gamma(1 + s) ds \text{ for } -1 < \gamma < 0
\end{aligned}$$

From the sum of the values of the various expressions, we obtain:

$$6.9984354800 + 0.23924625026 + 0.81679884826 + (20 \times 0.248754477033784 + 4 \ln(3) - 5/3 \ln(2) - 265/36) + 1/12(\ln(4\pi) - 0.5772156649 - 5/6) \times 10$$

Input interpretation

$$6.9984354800 + 0.23924625026 + 0.81679884826 + \\ \left(20 \times 0.248754477033784 + 4 \log(3) - \frac{5}{3} \log(2) - \frac{265}{36} \right) + \\ \frac{1}{12} \left(\log(4 \pi) - 0.5772156649 - \frac{5}{6} \right) \times 10$$

$\log(x)$ is the natural logarithm

Result

9.8413922358...

$$9.8413922358... \approx \pi^2$$

Alternative representations

$$6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \\ \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\ \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = 13.02957011919568 - \frac{5}{3} \log(a) \log_a(2) + \\ 4 \log(a) \log_a(3) + \frac{10}{12} \left(-0.577216 + \log(a) \log_a(4 \pi) - \frac{5}{6} \right) - \frac{265}{36}$$

$$6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \\ \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\ \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = 13.02957011919568 - \\ \frac{5 \log_e(2)}{3} + 4 \log_e(3) + \frac{10}{12} \left(-0.577216 + \log_e(4 \pi) - \frac{5}{6} \right) - \frac{265}{36}$$

$$6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \\ \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\ \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = 13.02957011919568 - \\ 4 \operatorname{Li}_1(-2) + \frac{5 \operatorname{Li}_1(-1)}{3} + \frac{10}{12} \left(-0.577216 - \operatorname{Li}_1(1 - 4 \pi) - \frac{5}{6} \right) - \frac{265}{36}$$

Series representations

$$\begin{aligned}
& 6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \\
& \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\
& \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = \\
& 4.493 - \frac{10}{3} i \pi \left[\frac{\arg(2-x)}{2 \pi} \right] + 8 i \pi \left[\frac{\arg(3-x)}{2 \pi} \right] + \frac{5}{3} i \pi \left[\frac{\arg(4 \pi-x)}{2 \pi} \right] + \\
& \frac{19 \log(x)}{6} + \sum_{k=1}^{\infty} \frac{(-1)^k (10(2-x)^k - 24(3-x)^k - 5(4 \pi-x)^k) x^{-k}}{6 k} \quad \text{for } x < 0
\end{aligned}$$

$$\begin{aligned}
& 6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \\
& \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\
& \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = \\
& 4.493 - \frac{5}{3} \left[\frac{\arg(2-z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + 4 \left[\frac{\arg(3-z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \\
& \frac{5}{6} \left[\frac{\arg(4 \pi-z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \frac{19 \log(z_0)}{6} - \frac{5}{3} \left[\frac{\arg(2-z_0)}{2 \pi} \right] \log(z_0) + \\
& 4 \left[\frac{\arg(3-z_0)}{2 \pi} \right] \log(z_0) + \frac{5}{6} \left[\frac{\arg(4 \pi-z_0)}{2 \pi} \right] \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k (10(2-z_0)^k - 24(3-z_0)^k - 5(4 \pi-z_0)^k) z_0^{-k}}{6 k}
\end{aligned}$$

$$\begin{aligned}
& 6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \\
& \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36} \right) + \\
& \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 = 4.493 - \\
& 3.33333 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + 8 i \pi \left[-\frac{-\pi + \arg\left(\frac{3}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + \\
& 1.66667 i \pi \left[-\frac{-\pi + \arg\left(\frac{4 \pi}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + 3.16667 \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k (1.66667(2-z_0)^k - 4(3-z_0)^k - 0.833333(4 \pi-z_0)^k) z_0^{-k}}{k}
\end{aligned}$$

Integral representations

$$6.99843548000000 + 0.239246250260000 + 0.816798848260000 +$$

$$\left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) +$$

$$\frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6}\right) 10 = 4.493 +$$

$$\int_1^{4\pi} \frac{2.5 + 6.66667 \pi^2 - 8.41667 t + 3.16667 t^2 + \pi (-8.33333 + 17.6667 t)}{t (-2 + 4 \pi + t) (-1.5 + 2 \pi + t)} dt$$

$$6.99843548000000 + 0.239246250260000 + 0.816798848260000 +$$

$$\left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{1}{3} \log(2) 5 - \frac{265}{36}\right) +$$

$$\frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6}\right) 10 =$$

$$4.493 + \int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{1}{i \pi \Gamma(1-s)} (-2+8\pi)^{-s} (0.416667 \times 2^s + 2(-1+4\pi)^s -$$

$$0.833333 (-2+8\pi)^s) \Gamma(-s)^2 \Gamma(1+s) ds \text{ for } -1 < \gamma < 0$$

From which:

$$1/6((6.9984354800+0.23924625026+0.81679884826+(20*0.248754477033784+4$$

$$\ln(3) - 5/3 \ln(2) - 265/36) + 1/12(\ln(4\pi)-0.5772156649-5/6)*10))$$

Input interpretation

$$\frac{1}{6} \left(6.9984354800 + 0.23924625026 + 0.81679884826 + \right.$$

$$\left(20 \times 0.248754477033784 + 4 \log(3) - \frac{5}{3} \log(2) - \frac{265}{36} \right) +$$

$$\left. \frac{1}{12} \left(\log(4 \pi) - 0.5772156649 - \frac{5}{6} \right) \times 10 \right)$$

$\log(x)$ is the natural logarithm

Result

1.6402320393...

$$1.6402320393 \dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$$

Alternative representations

$$\begin{aligned} & \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\ & \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\ & \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = \\ & \frac{1}{6} \left(13.02957011919568 - \frac{5}{3} \log(a) \log_a(2) + 4 \log(a) \log_a(3) + \right. \\ & \quad \left. \frac{10}{12} \left(-0.577216 + \log(a) \log_a(4 \pi) - \frac{5}{6} \right) - \frac{265}{36} \right) \end{aligned}$$

$$\begin{aligned} & \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\ & \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\ & \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = \\ & \frac{1}{6} \left(13.02957011919568 - \frac{5 \log_e(2)}{3} + 4 \log_e(3) + \right. \\ & \quad \left. \frac{10}{12} \left(-0.577216 + \log_e(4 \pi) - \frac{5}{6} \right) - \frac{265}{36} \right) \end{aligned}$$

$$\begin{aligned} & \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\ & \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\ & \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = \\ & \frac{1}{6} \left(13.02957011919568 - 4 \operatorname{Li}_1(-2) + \frac{5 \operatorname{Li}_1(-1)}{3} + \right. \\ & \quad \left. \frac{10}{12} \left(-0.577216 - \operatorname{Li}_1(1 - 4 \pi) - \frac{5}{6} \right) - \frac{265}{36} \right) \end{aligned}$$

Series representations

$$\begin{aligned} & \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\ & \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\ & \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = \\ & 0.748834 - \frac{5}{9} i \pi \left[\frac{\arg(2-x)}{2 \pi} \right] + \frac{4}{3} i \pi \left[\frac{\arg(3-x)}{2 \pi} \right] + \frac{5}{18} i \pi \left[\frac{\arg(4 \pi - x)}{2 \pi} \right] + \\ & \frac{19 \log(x)}{36} + \sum_{k=1}^{\infty} \frac{(-1)^k (10(2-x)^k - 24(3-x)^k - 5(4 \pi - x)^k) x^{-k}}{36 k} \quad \text{for } x < 0 \end{aligned}$$

$$\begin{aligned} & \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\ & \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\ & \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = \\ & 0.748834 - 0.555556 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + \\ & 1.33333 i \pi \left[-\frac{-\pi + \arg\left(\frac{3}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + \\ & 0.277778 i \pi \left[-\frac{-\pi + \arg\left(\frac{4 \pi}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + 0.527778 \log(z_0) + \\ & \sum_{k=1}^{\infty} \frac{(-1)^k (0.277778 (2 - z_0)^k - 0.666667 (3 - z_0)^k - 0.138889 (4 \pi - z_0)^k) z_0^{-k}}{k} \end{aligned}$$

$$\begin{aligned}
& \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\
& \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\
& \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = \\
& 0.748834 - \frac{5}{18} \left[\frac{\arg(2 - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \frac{2}{3} \left[\frac{\arg(3 - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \\
& \frac{5}{36} \left[\frac{\arg(4 \pi - z_0)}{2 \pi} \right] \log\left(\frac{1}{z_0}\right) + \frac{19 \log(z_0)}{36} - \frac{5}{18} \left[\frac{\arg(2 - z_0)}{2 \pi} \right] \log(z_0) + \\
& \frac{2}{3} \left[\frac{\arg(3 - z_0)}{2 \pi} \right] \log(z_0) + \frac{5}{36} \left[\frac{\arg(4 \pi - z_0)}{2 \pi} \right] \log(z_0) + \\
& \sum_{k=1}^{\infty} \frac{(-1)^k (10(2 - z_0)^k - 24(3 - z_0)^k - 5(4 \pi - z_0)^k) z_0^{-k}}{36 k}
\end{aligned}$$

Integral representations

$$\begin{aligned}
& \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\
& \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\
& \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = 0.748834 + \int_1^{4\pi} \\
& \frac{0.416667 + 1.11111 \pi^2 - 1.40278 t + 0.527778 t^2 + \pi(-1.38889 + 2.94444 t)}{t(-2 + 4 \pi + t)(-1.5 + 2 \pi + t)} \\
& dt
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{6} \left(6.99843548000000 + 0.239246250260000 + 0.816798848260000 + \right. \\
& \quad \left(20 \times 0.2487544770337840000 + 4 \log(3) - \frac{5 \log(2)}{3} - \frac{265}{36} \right) + \\
& \quad \left. \frac{1}{12} \left(\log(4 \pi) - 0.577216 - \frac{5}{6} \right) 10 \right) = \\
& 0.748834 + \int_{-i \infty + \gamma}^{i \infty + \gamma} \frac{1}{i \pi \Gamma(1 - s)} (-2 + 8 \pi)^{-s} \\
& (0.0694444 \times 2^s + 0.333333 (-1 + 4 \pi)^s - 0.138889 (-2 + 8 \pi)^s) \\
& \Gamma(-s)^2 \Gamma(1 + s) ds \text{ for } -1 < \gamma < 0
\end{aligned}$$

Now, we have that:

$$\frac{1}{(2\pi)^n} \int d^n k \frac{1}{(k^2 + \mu_0^2)^2} \quad (13.2)$$

in the limits $\mu_0 \rightarrow \infty$ and $n \rightarrow 4$. For definiteness, we specify the theory at $n \neq 4$ dimensions: All trivial factors $(2\pi)^4$ must also be replaced by $(2\pi)^n$, which leads to the factor $(2\pi)^{-n}$ in (13.2).

The integral (13.2) is in this limit

$$\frac{1}{(4\pi)^2} \left[\frac{2}{4-n} - \gamma - 2 \ln \mu_0 + \ln 4\pi + O(4-n) + O\left(\frac{1}{\mu_0^2}\right) \right], \quad (13.3)$$

where γ is again Euler's constant. So here

$$\ln \mu_0 \rightarrow \frac{1}{4-n} - \frac{1}{2} \gamma + \frac{1}{2} \ln 4\pi. \quad (13.4)$$

Let there be $N^s(t)$ scalar multiplets for each isospin t , where each complex scalar multiplet counts as one, and each real scalar multiplet counts as one-half. Then from (7.6) we obtain the total contribution from vector and scalar nonzero modes:

$$A = 6.9984354800 \quad N^s(t) = 1+1+1+1+1/2+1/2+1/2+1/2+1/2+1/2+1/2+1/2 = 8$$

$$A(t) = 0.23924625026 \quad C(t) = 1 \quad \rho = 1 \quad n \geq 0, \quad n = 2; \quad \mu = 16 \quad a_1 = 10$$

$$g_B(n) = \mu^{4-n} \left(g_R^D(\mu) + \frac{a_1}{n-4} + \cdots \right),$$

$$y = 16^2(16x-5)$$

Input

$$y = 16^2(16x-5)$$

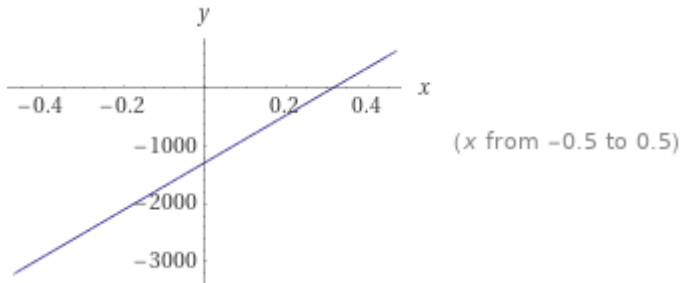
Result

$$y = 256(16x-5)$$

Geometric figure

line

Plot



Alternate form

$$-4096x + y + 1280 = 0$$

Expanded form

$$y = 4096x - 1280$$

Root

$$x = \frac{5}{16}$$

Properties as a real function

Domain

$\mathbb{R}$ (all real numbers)

Range

$\mathbb{R}$ (all real numbers)

Bijectivity

bijjective from its domain to $\mathbb{R}$

$\mathbb{R}$ is the set of real numbers

Partial derivatives

$$\frac{\partial}{\partial x}(256(16x - 5)) = 4096$$

$$\frac{\partial}{\partial y}(256(16x - 5)) = 0$$

We consider $x = 1/2$:

$$16^2(16 \cdot 1/2 - 5)$$

Input

$$16^2 \left(16 \times \frac{1}{2} - 5 \right)$$

Exact result

768

768 result very near to the Charged rho meson mass 775.11

$$A = 6.9984354800 \quad N^s(t) = 1+1+1+1+1/2+1/2+1/2+1/2+1/2+1/2+1/2+1/2 = 8$$

$$A(t) = 0.23924625026 \quad C(t) = 1 \quad \rho = 1 \quad n \geq 0, \quad n = 2; \quad \mu = 16; \quad a_1 = 10$$

$$g_B^2(n) = 768^2 \quad g = 0.30282212$$

From:

$$2^{14} \pi^6 g^{-8} \frac{d^4 z d\rho}{\rho^5} \exp \left\{ -\frac{8\pi^2}{g^2(\mu_0)} + \ln(\mu_0 \rho) \left[\frac{22}{3} - \frac{1}{6} \sum_t N^s(t) C(t) \right] - \alpha(1) - \sum_t N^s(t) \alpha(t) \right\}.$$

$$2^{14} \pi^6 (0.30282212)^{-8}$$

Input interpretation

$$\frac{2^{14} \pi^6}{0.30282212^8}$$

Result

$$2.227504... \times 10^{11}$$

$$2.227504... * 10^{11}$$

$$\exp\left[\frac{-8\pi^2}{(0.30282212^2 \times 16)} + \ln(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.248754477033784 + \frac{1}{3} \ln(2) - \frac{16}{9}\right) - 8 \left(2 \times 0.248754477033784 - \frac{1}{6} \ln(2) - \frac{17}{72}\right)\right]$$

Input interpretation

$$\exp\left(\frac{-8\pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.248754477033784 + \frac{1}{3} \log(2) - \frac{16}{9}\right) - 8 \left(2 \times 0.248754477033784 - \frac{1}{6} \log(2) - \frac{17}{72}\right)\right)$$

$\log(x)$ is the natural logarithm

Result

$$1.42657... \times 10^{-17}$$

$$1.42657... * 10^{-17}$$

$$2^{14} \pi^6 (0.30282212)^{-8} \left(\left(\exp\left(\frac{-8\pi^2}{(0.30282212)^2 \times 16}\right) + \ln(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.248754477033784 + \frac{1}{3} \ln(2) - \frac{16}{9}\right) - 8 \left(2 \times 0.248754477033784 - \frac{1}{6} \ln(2) - \frac{17}{72}\right) \right) \right)$$

Input interpretation

$$\frac{1}{0.30282212^8} 2^{14} \pi^6 \exp\left(\frac{-8\pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.248754477033784 + \frac{1}{3} \log(2) - \frac{16}{9}\right) - 8 \left(2 \times 0.248754477033784 - \frac{1}{6} \log(2) - \frac{17}{72}\right)\right)$$

$\log(x)$ is the natural logarithm

Result

$$3.17769... \times 10^{-6}$$

$$3.17769... \times 10^{-6}$$

From which:

$$\text{colog}\left(\left(\left(2^{14} \pi^6 (0.30282212)^{-8} \left(\left(\exp\left(\frac{-8\pi^2}{(0.30282212)^2 \times 16}\right) + \ln(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.248754477033784 + \frac{1}{3} \ln(2) - \frac{16}{9}\right) - 8 \left(2 \times 0.248754477033784 - \frac{1}{6} \ln(2) - \frac{17}{72}\right) \right) \right) \right)\right)\right)$$

Input interpretation

$$-\log\left(\frac{1}{0.30282212^8} 2^{14} \pi^6 \exp\left(\frac{-8\pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1\right) - \left(8 \times 0.248754477033784 + \frac{1}{3} \log(2) - \frac{16}{9}\right) - 8 \left(2 \times 0.248754477033784 - \frac{1}{6} \log(2) - \frac{17}{72}\right)\right)\right)$$

$\log(x)$ is the natural logarithm

Result

$$12.65936...$$

$$12.65936.... \text{ very near to the black hole entropy BH } 12.5664$$

Alternative representations

$$\begin{aligned}
 & -\log\left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \right.\right. \\
 & \quad \left.\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \right. \\
 & \quad \left.\left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right)\right) = -\log_e\left(\frac{1}{0.302822^8}\right. \\
 & \quad \exp\left(-1.990035816270272000 - \frac{\log(2)}{3} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) + \frac{16}{9} - \right. \\
 & \quad \left.\left. 8\left(0.4975089540675680000 - \frac{\log(2)}{6} - \frac{17}{72}\right) - \frac{8\pi^2}{16 \times 0.302822^2}\right) 2^{14} \pi^6\right)
 \end{aligned}$$

$$\begin{aligned}
 & -\log\left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \right.\right. \\
 & \quad \left.\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \right. \\
 & \quad \left.\left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right)\right) = -\log(a) \log_a\left(\right. \\
 & \quad \frac{1}{0.302822^8} \exp\left(-1.990035816270272000 - \frac{\log(2)}{3} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) + \frac{16}{9} - \right. \\
 & \quad \left.\left. 8\left(0.4975089540675680000 - \frac{\log(2)}{6} - \frac{17}{72}\right) - \frac{8\pi^2}{16 \times 0.302822^2}\right) 2^{14} \pi^6\right)
 \end{aligned}$$

Series representations

$$\begin{aligned}
 & -\log\left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \right.\right. \\
 & \quad \left.\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \right. \\
 & \quad \left.\left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right)\right) = \\
 & \quad \sum_{k=1}^{\infty} \frac{(-1)^k \left(-1 + 2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))\right)^k}{k}
 \end{aligned}$$

$$\begin{aligned}
& -\log\left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \right.\right. \\
& \quad \left.\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \right. \\
& \quad \left.\left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right)\right) = \\
& -2i\pi \left[\frac{1}{2\pi} \arg(-x + 2.31696 \times 10^8 \pi^6 \right. \\
& \quad \left. \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))) \right] - \\
& \log(x) + \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k x^{-k} (-x + 2.31696 \times 10^8 \pi^6 \\
& \quad \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)))^k \text{ for } x < 0
\end{aligned}$$

$$\begin{aligned}
& -\log\left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \right.\right. \\
& \quad \left.\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \right. \\
& \quad \left.\left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right)\right) = \\
& -2i\pi \left[-\frac{-\pi + \arg\left(\frac{2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))}{z_0}\right) + \arg(z_0)}{2\pi} \right] - \\
& \log(z_0) + \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k \\
& \quad \frac{(2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)) - z_0)^k}{z_0^{-k}}
\end{aligned}$$

Integral representation

$$\begin{aligned}
& -\log\left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \right.\right. \\
& \quad \left.\left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \right. \\
& \quad \left.\left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right)\right) = \\
& -\int_1^{2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))} \frac{1}{t} dt
\end{aligned}$$

Alternative representations

$$\begin{aligned} & \frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \right. \\ & \quad \log(16) \left(\frac{22}{3} - \frac{8}{6}\right) - \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \\ & \quad \left. 8 \left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right) = \frac{1}{0.302822^8} \\ & \exp\left(-1.990035816270272000 - \frac{1}{3} \log(a) \log_a(2) + \log(a) \log_a(16) \left(\frac{22}{3} - \frac{8}{6}\right) + \right. \\ & \quad \frac{16}{9} - 8 \left(0.4975089540675680000 - \frac{1}{6} \log(a) \log_a(2) - \frac{17}{72}\right) - \\ & \quad \left. \frac{8\pi^2}{16 \times 0.302822^2}\right) 2^{14} \pi^6 \end{aligned}$$

$$\begin{aligned} & \frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \right. \\ & \quad \log(16) \left(\frac{22}{3} - \frac{8}{6}\right) - \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \\ & \quad \left. 8 \left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right) = \frac{1}{0.302822^8} \\ & \exp\left(-1.990035816270272000 - \frac{\log_e(2)}{3} + \log_e(16) \left(\frac{22}{3} - \frac{8}{6}\right) + \frac{16}{9} - \right. \\ & \quad \left. 8 \left(0.4975089540675680000 - \frac{\log_e(2)}{6} - \frac{17}{72}\right) - \frac{8\pi^2}{16 \times 0.302822^2}\right) 2^{14} \pi^6 \end{aligned}$$

Series representations

$$\begin{aligned} & \frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \right. \\ & \quad \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \\ & \quad \left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right) = 2.31696 \times 10^8 \pi^6 \\ & \exp\left(-2.30344 - 5.45249 \pi^2 + 2i\pi \left[\frac{\arg(2-x)}{2\pi}\right] + 12i\pi \left[\frac{\arg(16-x)}{2\pi}\right] + \right. \\ & \quad \left. 7. \log(x) + \sum_{k=1}^{\infty} \frac{(-1)^k \left(-(2-x)^k - 6(16-x)^k\right) x^{-k}}{k}\right) \text{ for } x < 0 \end{aligned}$$

$$\begin{aligned} & \frac{1}{0.302822^8} 2^{14} \pi^6 \exp\left(-\frac{8\pi^2}{0.302822^2 \times 16} + \right. \\ & \quad \left. \log(16)\left(\frac{22}{3} - \frac{8}{6}\right) - \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9}\right) - \right. \\ & \quad \left. 8\left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72}\right)\right) = \\ & 2.31696 \times 10^8 \pi^6 \exp\left(-2.30344 - 5.45249 \pi^2 + 7. \log(z_0) + \right. \\ & \quad \left. \left[\frac{\arg(2-z_0)}{2\pi}\right] \left(\log\left(\frac{1}{z_0}\right) + \log(z_0)\right) + \left[\frac{\arg(16-z_0)}{2\pi}\right] \left(6 \log\left(\frac{1}{z_0}\right) + 6 \log(z_0)\right) + \right. \\ & \quad \left. \sum_{k=1}^{\infty} \frac{(-1)^k \left(-(2-z_0)^k - 6(16-z_0)^k\right) z_0^{-k}}{k}\right) \end{aligned}$$

$$\begin{aligned}
& \frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \right. \\
& \quad \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \\
& \quad \left. 8 \left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) = \\
& 2.31696 \times 10^8 \pi^6 \exp \left(-2.30344 - 5.45249 \pi^2 + 2 i \pi \left[-\frac{-\pi + \arg\left(\frac{2}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + \right. \\
& \quad 12 i \pi \left[-\frac{-\pi + \arg\left(\frac{16}{z_0}\right) + \arg(z_0)}{2 \pi} \right] + 7 \cdot \log(z_0) + \\
& \quad \left. \sum_{k=1}^{\infty} \frac{(-1)^k \left(-(2 - z_0)^k - 6(16 - z_0)^k \right) z_0^{-k}}{k} \right)
\end{aligned}$$

Integral representation

$$\begin{aligned}
& \frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \\
& \quad \left(8 \times 0.2487544770337840000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \\
& \quad \left. 8 \left(2 \times 0.2487544770337840000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) = 2.31696 \times 10^8 \pi^6 \\
& \exp \left(\frac{i \pi \left(-2.30344 - 5.45249 \pi^2 \right) + \int_{-i \infty + \gamma}^{i \infty + \gamma} \frac{0.5 \times 15^{-s} (6 + 15^s) \Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds}{i \pi} \right) \\
& \text{for } -1 < \gamma < 0
\end{aligned}$$

144*(colog((((2^14*Pi^6*(0.30282212)^-8 (((exp[(-
8Pi^2)/(0.30282212^2*16)+ln(16) (22/3-1/6*8*1)- (8*0.2487544770337+1/3 ln(2) –
16/9)-8*((2*0.2487544770337-1/6 ln(2) – 17/72)))))))))))+(1-1.65578))

Input interpretation

$$144 \left(-\log \left(\frac{1}{0.30282212^8} 2^{14} \pi^6 \exp \left(\frac{-8 \pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 \right) - \right. \right. \right. \\ \left. \left. \left(8 \times 0.2487544770337 + \frac{1}{3} \log(2) - \frac{16}{9} \right) - \right. \right. \\ \left. \left. \left. 8 \left(2 \times 0.2487544770337 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) \right) \right) + (1 - 1.65578) \right)$$

$\log(x)$ is the natural logarithm

Result

1728.515...

1728.515....

This result is very near to the mass of candidate glueball **$f_0(1710)$ scalar meson**. Furthermore, 1728 occurs in the algebraic formula for the j-invariant of an elliptic curve. ($1728 = 8^2 * 3^3$) The number 1728 is one less than the Hardy–Ramanujan number 1729 (taxicab number)

Alternative representations

$$144 \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \\ \left. \left. \left(8 \times 0.24875447703370000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \right. \\ \left. \left. \left. 8 \left(2 \times 0.24875447703370000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) \right) + (1 - 1.65578) \right) = \\ 144 \left(-0.65578 - \log_e \left(\frac{1}{0.302822^8} \exp \left(-1.9900358162696000 - \frac{\log(2)}{3} + \log(16) \right. \right. \right. \\ \left. \left. \left(\frac{22}{3} - \frac{8}{6} \right) + \frac{16}{9} - 8 \left(0.49750895406740000 - \frac{\log(2)}{6} - \frac{17}{72} \right) - \right. \right. \\ \left. \left. \left. \frac{8 \pi^2}{16 \times 0.302822^2} \right) 2^{14} \pi^6 \right) \right)$$

$$\begin{aligned}
& 144 \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \\
& \quad \left. \left(8 \times 0.24875447703370000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.24875447703370000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) \right) + \\
& (1 - 1.65578) = 144 \left(-0.65578 - \log(a) \log_a \left(\frac{1}{0.302822^8} \right. \right. \\
& \quad \exp \left(-1.9900358162696000 - \frac{\log(2)}{3} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) + \right. \\
& \quad \left. \frac{16}{9} - 8 \left(0.49750895406740000 - \frac{\log(2)}{6} - \frac{17}{72} \right) - \right. \\
& \quad \left. \left. \left. \frac{8 \pi^2}{16 \times 0.302822^2} \right) 2^{14} \pi^6 \right) \right)
\end{aligned}$$

Series representations

$$\begin{aligned}
& 144 \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \\
& \quad \left. \left(8 \times 0.24875447703370000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.24875447703370000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) \right) + \\
& (1 - 1.65578) = -94.4323 + 144 \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k \\
& \quad (-1 + 2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)))^k
\end{aligned}$$

$$\begin{aligned}
& 144 \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \\
& \quad \left. \left(8 \times 0.24875447703370000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.24875447703370000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) \right) + \\
& \quad (1 - 1.65578) \Bigg) = -94.4323 - 288 i \pi \left[\frac{1}{2 \pi} \arg(-x + \right. \\
& \quad \left. 2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))) \right] - \\
& \quad 144 \log(x) + 144 \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k x^{-k} (-x + 2.31696 \times 10^8 \pi^6 \\
& \quad \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)))^k \text{ for } x < 0
\end{aligned}$$

$$\begin{aligned}
& 144 \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \\
& \quad \left. \left(8 \times 0.24875447703370000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.24875447703370000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) \right) + \\
& \quad (1 - 1.65578) \Bigg) = -94.4323 - 288 i \pi \\
& \quad \left[-\frac{-\pi + \arg \left(\frac{2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))}{z_0} \right) + \arg(z_0)}{2 \pi} \right] - \\
& \quad 144 \log(z_0) + \\
& \quad 144 \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k \\
& \quad \left(\frac{2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))}{z_0} - z_0 \right)^k
\end{aligned}$$

Integral representation

$$144 \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.24875447703370000 + \frac{\log(2)}{3} - \frac{16}{9} \right) - 8 \left(2 \times 0.24875447703370000 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) = -94.4323 - 144 \int_1^{2.31696 \times 10^8} \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)) \frac{1}{t} dt$$

$$[144 * (\text{colog}((((2^{14} * \pi^6 * (0.30282212)^{-8} ((\exp[(-8\pi^2)/(0.30282212^2 * 16) + \ln(16) (22/3 - 1/6 * 8 * 1) - (8 * 0.248754477 + 1/3 \ln(2) - 16/9) - 8 * ((2 * 0.248754477 - 1/6 \ln(2) - 17/72))))))))) + (1 - 1.65578)))]^{1/15}$$

Input interpretation

$$\left(144 \left(-\log \left(\frac{1}{0.30282212^8} 2^{14} \pi^6 \exp \left(\frac{-8\pi^2}{0.30282212^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 \right) - \left(8 \times 0.248754477 + \frac{1}{3} \log(2) - \frac{16}{9} \right) - 8 \left(2 \times 0.248754477 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^{(1/15)}$$

$\log(x)$ is the natural logarithm

Result

1.6437844723163222103846531080814471775531668323567830776004937065

...

$$1.6437844723 \dots \approx \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$$

We obtain also:

$$(1/27(144(\text{colog}((((2^{14} * \pi^6 * (0.30282212)^{-8} ((\exp[(-8\pi^2)/(0.30282212^2 * 16) + \ln(16) (22/3 - 1/6 * 8 * 1) - (8 * 0.24875447 + 1/3 \ln(2) - 16/9) - 8 * ((2 * 0.24875447 - 1/6 \ln(2) - 17/72))))))))) + (1 - 1.65578)))]^2 + 276 - 5/2$$

Input interpretation

$$\left(\frac{1}{27} \left(144 \left(-\log \left(\frac{1}{0.30282212^8} 2^{14} \pi^6 \exp \left(\frac{-8 \pi^2}{0.30282212^2 \times 16} + \right. \right. \right. \right. \right. \right. \\ \left. \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 \right) - \left(8 \times 0.24875447 + \frac{1}{3} \log(2) - \right. \right. \\ \left. \left. \frac{16}{9} \right) - 8 \left(2 \times 0.24875447 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) \right) + \\ \left. \left. \left. \left. \left. (1 - 1.65578) \right) \right) \right) \right)^2 + 276 - \frac{5}{2}$$

$\log(x)$ is the natural logarithm

Result

4371.94...

$$4371.94 \approx 4372 = 64^2 + 276$$

Alternative representations

$$\left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \right. \right. \right. \right. \right. \right. \\ \left. \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \\ \left. \left. 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 + \\ 276 - \frac{5}{2} = \frac{547}{2} + \left(\frac{144}{27} \left(-0.65578 - \log_e \left(\frac{1}{0.302822^8} \right. \right. \right. \right. \\ \left. \exp \left(-1.99004 - \frac{\log(2)}{3} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) + \frac{16}{9} - \right. \right. \\ \left. \left. 8 \left(0.497509 - \frac{\log(2)}{6} - \frac{17}{72} \right) - \frac{8 \pi^2}{16 \times 0.302822^2} \right) 2^{14} \pi^6 \right) \right) \right)^2$$

$$\begin{aligned}
& \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \right. \right. \right. \right. \\
& \quad \left. \left. \left. \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 + \\
& 276 - \frac{5}{2} = \frac{547}{2} + \left(\frac{144}{27} \left(-0.65578 - \log(a) \log_a \left(\frac{1}{0.302822^8} \right. \right. \right. \right. \\
& \quad \left. \left. \left. \exp \left(-1.99004 - \frac{\log(2)}{3} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) + \frac{16}{9} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. 8 \left(0.497509 - \frac{\log(2)}{6} - \frac{17}{72} \right) - \frac{8\pi^2}{16 \times 0.302822^2} \right) 2^{14} \pi^6 \right) \right) \right)^2
\end{aligned}$$

Series representations

$$\begin{aligned}
& \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \right. \right. \right. \right. \\
& \quad \left. \left. \left. \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 + \\
& 276 - \frac{5}{2} = \frac{547}{2} + 28.4444 \left(0.65578 - \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k (-1 + 2.31696 \times 10^8 \right. \\
& \quad \left. \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)))^k \right)^2
\end{aligned}$$

$$\begin{aligned}
& \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \right. \\
& \quad \left. \left. \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - 8 \left(2 \times 0.248754 - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 + 276 - \frac{5}{2} = \\
& \frac{547}{2} + 113.778 \left(0.32789 + i\pi \left[\frac{1}{2\pi} \arg(-x + 2.31696 \times 10^8 \pi^6 \exp(-2.30344 - \right. \right. \\
& \quad \left. \left. 5.45249 \pi^2 + \log(2) + 6 \log(16))) \right] + 0.5 \log(x) - \right. \\
& \quad \left. 0.5 \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k x^{-k} (-x + 2.31696 \times 10^8 \pi^6 \exp(-2.30344 - \right. \\
& \quad \left. \left. 5.45249 \pi^2 + \log(2) + 6 \log(16)))^k \right)^2 \right) \text{ for } x < 0
\end{aligned}$$

$$\begin{aligned}
& \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \right. \right. \right. \right. \\
& \quad \left. \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \\
& \quad \left. \left. 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 + \\
& 276 - \frac{5}{2} = \frac{547}{2} + \frac{256}{9} \left(0.65578 + \log(z_0) + \left[\frac{1}{2\pi} \arg(2.31696 \times 10^8 \pi^6 \right. \right. \\
& \quad \left. \left. \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)) - z_0) \right] \right. \\
& \quad \left(\log \left(\frac{1}{z_0} \right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k (2.31696 \times 10^8 \pi^6 \exp(-2.30344 - \\
& \quad \left. \left. 5.45249 \pi^2 + \log(2) + 6 \log(16)) - z_0)^k z_0^{-k} \right)^2 \right)
\end{aligned}$$

Integral representation

$$\left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right)^2 + 276 - \frac{5}{2} = \frac{547}{2} + 28.4444 \right. \\ \left. \left(0.65578 + \int_1^{2.31696 \times 10^8} \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)) \frac{1}{t} dt \right)^2 \right)$$

$$2 \left(\frac{1}{27} \left(144 \left(\text{colog} \left(\left((2^{14} \pi^6 (0.302822^{12})^{-8} \left(\exp \left(\left(-\frac{8\pi^2}{(0.302822^{12})^2 \times 16} + \ln(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 \right) - (8 \times 0.24875447 + \frac{1}{3} \ln(2) - \frac{16}{9}) - 8 \left(2 \times 0.24875447 - \frac{1}{6} \ln(2) - \frac{17}{72} \right) \right) \right) \right) \right) \right) + (1 - 1.65578) \right) \right)^2 - 5 \right)$$

Input interpretation

$$2 \left(\frac{1}{27} \left(144 \left(-\log \left(\frac{1}{0.302822^{12}^8} 2^{14} \pi^6 \exp \left(\frac{-8\pi^2}{0.302822^{12}^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{1}{6} \times 8 \times 1 \right) - \left(8 \times 0.24875447 + \frac{1}{3} \log(2) - \frac{16}{9} \right) - 8 \left(2 \times 0.24875447 - \frac{1}{6} \log(2) - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 - 5 \right)$$

$\log(x)$ is the natural logarithm

Result

8191.88...

$$8191.88... \approx 8192$$

The total amplitude vanishes for gauge group $SO(8192)$, while the vacuum energy is negative and independent of the gauge group.

The vacuum energy and dilaton tadpole to lowest non-trivial order for the open bosonic string. While the vacuum energy is non-zero and independent of the gauge group, the dilaton tadpole is zero for a unique choice of gauge group, $SO(2^{13})$ i.e. $SO(8192)$. (From: “Dilaton Tadpole for the Open Bosonic String “ Michael R. Douglas and Benjamin Grinstein - September 2,1986)

Alternative representations

$$2 \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) + (1 - 1.65578) \right) \right)^2 - 5 = -5 + 2 \left(\frac{144}{27} \left(-0.65578 - \log_e \left(\frac{1}{0.302822^8} \exp \left(-1.99004 - \frac{\log(2)}{3} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) + \frac{16}{9} - 8 \left(0.497509 - \frac{\log(2)}{6} - \frac{17}{72} \right) - \frac{8\pi^2}{16 \times 0.302822^2} \right) 2^{14} \pi^6 \right) \right) \right)^2$$

$$2 \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8\pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) + (1 - 1.65578) \right) \right)^2 - 5 = -5 + 2 \left(\frac{144}{27} \left(-0.65578 - \log(a) \log_a \left(\frac{1}{0.302822^8} \exp \left(-1.99004 - \frac{\log(2)}{3} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) + \frac{16}{9} - 8 \left(0.497509 - \frac{\log(2)}{6} - \frac{17}{72} \right) - \frac{8\pi^2}{16 \times 0.302822^2} \right) 2^{14} \pi^6 \right) \right) \right)^2$$

Series representations

$$\begin{aligned}
& 2 \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \right. \right. \right. \right. \\
& \quad \left. \left. \left. \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 - \\
& 5 = -5 + 56.8889 \left(0.65578 - \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k (-1 + 2.31696 \times 10^8 \pi^6 \right. \\
& \quad \left. \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)))^k \right)^2
\end{aligned}$$

$$\begin{aligned}
& 2 \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \right. \right. \\
& \quad \left. \left. \left. \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - 8 \left(2 \times 0.248754 - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 - 5 = \\
& -5 + 227.556 \left(0.32789 + i \pi \left[\frac{1}{2 \pi} \arg(-x + 2.31696 \times 10^8 \pi^6 \exp(-2.30344 - \right. \right. \right. \\
& \quad \left. \left. \left. 5.45249 \pi^2 + \log(2) + 6 \log(16))) \right] + 0.5 \log(x) - \right. \right. \\
& \quad \left. 0.5 \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k x^{-k} (-x + 2.31696 \times 10^8 \pi^6 \exp(-2.30344 - \right. \\
& \quad \left. \left. 5.45249 \pi^2 + \log(2) + 6 \log(16)))^k \right)^2 \right) \text{ for } x < 0
\end{aligned}$$

$$\begin{aligned}
& 2 \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \right. \right. \right. \right. \\
& \quad \left. \left. \left. \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. 8 \left(2 \times 0.248754 - \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 - \\
& 5 = -5 + \frac{512}{9} \left(0.65578 + \log(z_0) + \left\lfloor \frac{1}{2\pi} \arg(2.31696 \times 10^8 \pi^6 \right. \right. \\
& \quad \left. \left. \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16)) - z_0 \right) \right\rfloor \\
& \quad \left(\log \left(\frac{1}{z_0} \right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{1}{k} (-1)^k (2.31696 \times 10^8 \pi^6 \exp(-2.30344 - \\
& \quad 5.45249 \pi^2 + \log(2) + 6 \log(16)) - z_0)^k z_0^{-k} \Big)^2
\end{aligned}$$

Integral representation

$$\begin{aligned}
& 2 \left(\frac{144}{27} \left(-\log \left(\frac{1}{0.302822^8} 2^{14} \pi^6 \exp \left(-\frac{8 \pi^2}{0.302822^2 \times 16} + \log(16) \left(\frac{22}{3} - \frac{8}{6} \right) - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \left(8 \times 0.248754 + \frac{\log(2)}{3} - \frac{16}{9} \right) - 8 \left(2 \times 0.248754 - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{\log(2)}{6} - \frac{17}{72} \right) \right) \right) + (1 - 1.65578) \right) \right)^2 - 5 = \\
& -5 + 56.8889 \left(0.65578 + \int_1^{2.31696 \times 10^8 \pi^6 \exp(-2.30344 - 5.45249 \pi^2 + \log(2) + 6 \log(16))} \frac{1}{t} dt \right)^2
\end{aligned}$$

Observations

We note that, from the number 8, we obtain as follows:

$$8^2$$

$$64$$

$$8^2 \times 2 \times 8$$

$$1024$$

$$8^4 = 8^2 \times 2^6$$

True

$$8^4 = 4096$$

$$8^2 \times 2^6 = 4096$$

$$2^{13} = 2 \times 8^4$$

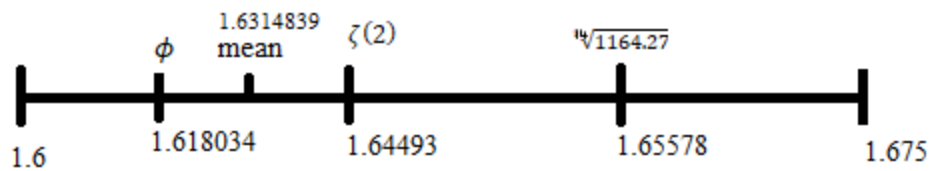
True

$$2^{13} = 8192$$

$$2 \times 8^4 = 8192$$

We notice how from the numbers 8 and 2 we get 64, 1024, 4096 and 8192, and that 8 is the fundamental number. In fact $8^2 = 64$, $8^3 = 512$, $8^4 = 4096$. We define it "fundamental number", since 8 is a Fibonacci number, which by rule, divided by the previous one, which is 5, gives 1.6, a value that tends to the golden ratio, as for all numbers in the Fibonacci sequence

“Golden” Range



Finally we note how $8^2 = 64$, multiplied by 27, to which we add 1, is equal to 1729, the so-called "Hardy-Ramanujan number". Then taking the 15th root of 1729, we obtain a value close to $\zeta(2)$ that 1.6438 ..., which, in turn, is included in the range of what we call "golden numbers"

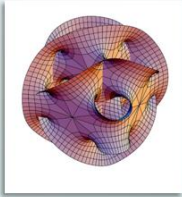
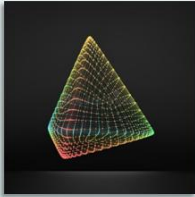
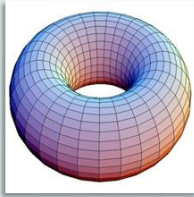
Furthermore for all the results very near to 1728 or 1729, adding $64 = 8^2$, one obtain values about equal to 1792 or 1793. These are values almost equal to the Planck multipole spectrum frequency 1792.35 and to the hypothetical Gluino mass

Appendix

Outlook

Remarkably rich (apparently **UNIQUE**) framework

BUT :



Why a given **“shape” of the extra dimensions** ?
[**CRUCIAL**, it determines the predictions for α , ...]

✎ ✂ ➦ ➤A. Sagnotti – AstronomiAmo, 23.4.202021

From: A. Sagnotti – AstronomiAmo, 23.04.2020

In the above figure, it is said that: “why a given shape of the extra dimensions? Crucial, it determines the predictions for α ”.

We propose that whatever shape the compactified dimensions are, their geometry must be based on the values of the golden ratio and $\zeta(2)$, (the latter connected to 1728 or 1729, whose fifteenth root provides an excellent approximation to the above mentioned value) which are recurrent as solutions of the equations that we are going to develop. It is important to specify that the initial conditions are **always** values belonging to a fundamental chapter of the work of S. Ramanujan "Modular equations and Approximations to Pi" (see references). These values are some multiples of 8 (64 and 4096), 276, which added to 4096, is equal to 4372, and finally $e^{\pi\sqrt{22}}$

We have, in certain cases, the following connections:

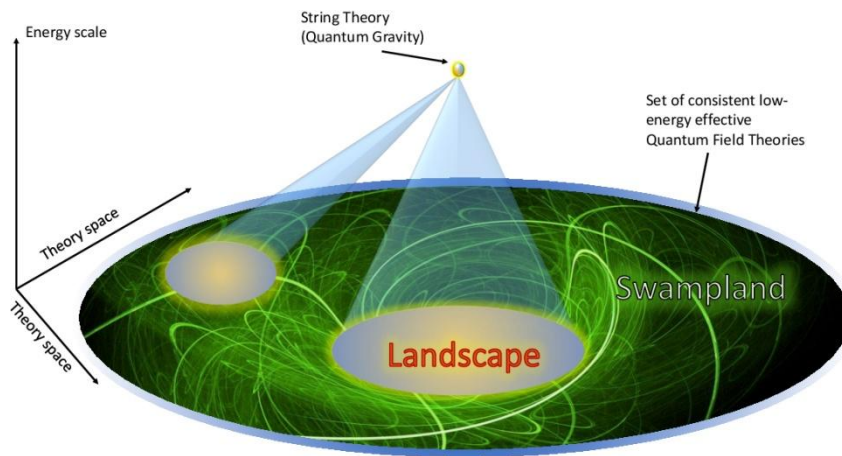
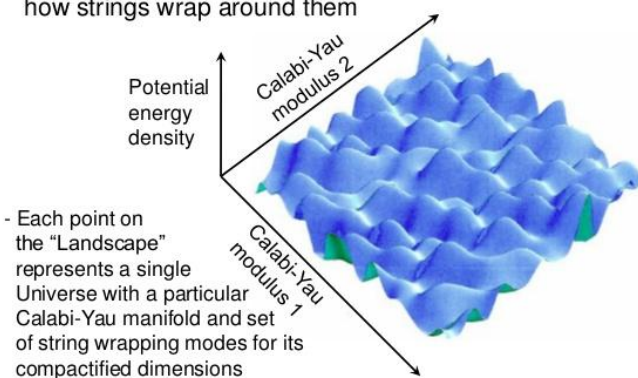


Fig. 1

The String Theory “Landscape”

- Graph axes show only 2 out of hundreds of parameters (“moduli”) that determine the exact Calabi-Yau manifolds and how strings wrap around them



- Each point on the “Landscape” represents a single Universe with a particular Calabi-Yau manifold and set of string wrapping modes for its compactified dimensions
- Each Universe could be realized in a separate post-inflation “bubble”

Fig. 2

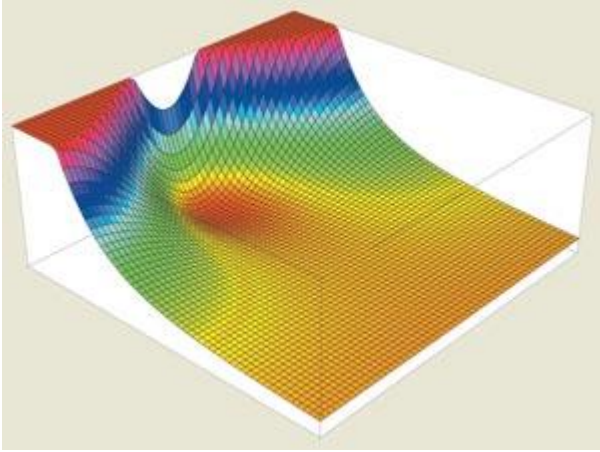


Fig. 3

Stringscape - a small part of the string-theory landscape showing the new de Sitter solution as a local minimum of the energy (vertical axis). The global minimum occurs at the infinite size of the extra dimensions on the extreme right of the figure.

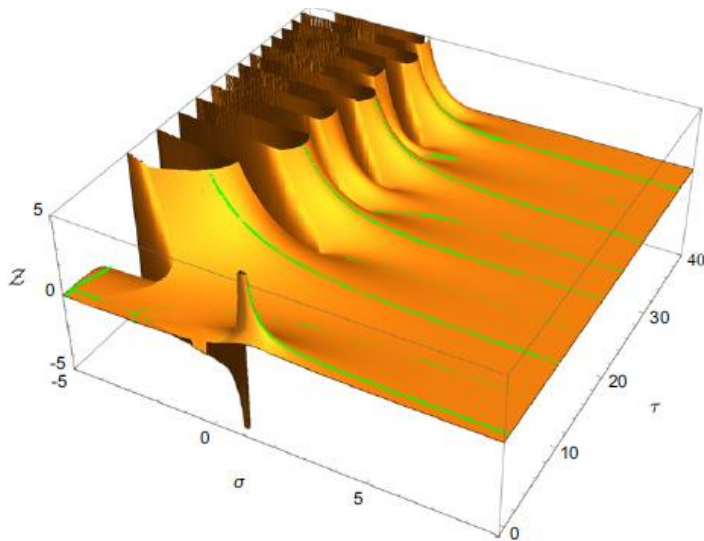


Figure 2. Lines in the complex plane where the Riemann zeta function ζ is real (green) depicted on a relief representing the positive absolute value of ζ for arguments $s \equiv \sigma + i\tau$ where the real part of ζ is positive, and the negative absolute value of ζ where the real part of ζ is negative. This representation brings out most clearly that the lines of constant phase corresponding to phases of integer multiples of 2π run down the hills on the left-hand side, turn around on the right and terminate in the non-trivial zeros. This pattern repeats itself infinitely many times. The points of arrival and departure on the right-hand side of the picture are equally spaced and given by equation (11).

Fig. 4

With regard the Fig. 4 the points of arrival and departure on the right-hand side of the picture are equally spaced and given by the following equation:

$$\tau'_k \equiv k \frac{\pi}{\ln 2},$$

with $k = \dots, -2, -1, 0, 1, 2, \dots$

we obtain:

$$2\pi/(\ln(2))$$

Input:

$$2 \times \frac{\pi}{\log(2)}$$

Exact result:

$$\frac{2\pi}{\log(2)}$$

Decimal approximation:

9.0647202836543876192553658914333336203437229354475911683720330958
...

9.06472028365....

Alternative representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log_e(2)}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log(a) \log_a(2)}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2 \coth^{-1}(3)}$$

Series representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2i\pi \left\lfloor \frac{\arg(2-x)}{2\pi} \right\rfloor + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-x)^k x^{-k}}{k}} \quad \text{for } x < 0$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\log(z_0) + \left\lfloor \frac{\arg(2-z_0)}{2\pi} \right\rfloor \left(\log\left(\frac{1}{z_0}\right) + \log(z_0) \right) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k}}$$

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{2i\pi \left\lfloor \frac{\pi - \arg\left(\frac{1}{z_0}\right) - \arg(z_0)}{2\pi} \right\rfloor + \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (2-z_0)^k z_0^{-k}}{k}}$$

Integral representations:

$$\frac{2\pi}{\log(2)} = \frac{2\pi}{\int_1^2 \frac{1}{t} dt}$$

$$\frac{2\pi}{\log(2)} = \frac{4i\pi^2}{\int_{-i\infty+\gamma}^{i\infty+\gamma} \frac{\Gamma(-s)^2 \Gamma(1+s)}{\Gamma(1-s)} ds} \quad \text{for } -1 < \gamma < 0$$

From which:

$$(2\pi/(\ln(2))) \cdot (1/12 \pi \log(2))$$

Input:

$$\left(2 \times \frac{\pi}{\log(2)}\right) \left(\frac{1}{12} \pi \log(2)\right)$$

$\log(x)$ is the natural logarithm

Exact result:

$$\frac{\pi^2}{6}$$

Decimal approximation:

1.6449340668482264364724151666460251892189499012067984377355582293
...

$$1.6449340668\dots = \zeta(2) = \frac{\pi^2}{6} = 1.644934 \dots$$

Mathematical connections with some sectors of String Theory

From:

Modular equations and approximations to π - *Srinivasa Ramanujan*

Quarterly Journal of Mathematics, XLV, 1914, 350 – 372

We have that:

Hence

$$\begin{aligned} 64g_{22}^{24} &= e^{\pi\sqrt{22}} - 24 + 276e^{-\pi\sqrt{22}} - \dots, \\ 64g_{22}^{-24} &= 4096e^{-\pi\sqrt{22}} + \dots, \end{aligned}$$

so that

$$64(g_{22}^{24} + g_{22}^{-24}) = e^{\pi\sqrt{22}} - 24 + 4372e^{-\pi\sqrt{22}} + \dots = 64\{(1 + \sqrt{2})^{12} + (1 - \sqrt{2})^{12}\}.$$

Hence

$$e^{\pi\sqrt{22}} = 2508951.9982\dots$$

Again

$$G_{37} = (6 + \sqrt{37})^{\frac{1}{4}},$$

$$\begin{aligned} 64G_{37}^{24} &= e^{\pi\sqrt{37}} + 24 + 276e^{-\pi\sqrt{37}} + \dots, \\ 64G_{37}^{-24} &= 4096e^{-\pi\sqrt{37}} - \dots, \end{aligned}$$

so that

$$64(G_{37}^{24} + G_{37}^{-24}) = e^{\pi\sqrt{37}} + 24 + 4372e^{-\pi\sqrt{37}} - \dots = 64\{(6 + \sqrt{37})^6 + (6 - \sqrt{37})^6\}.$$

Hence

$$e^{\pi\sqrt{37}} = 199148647.999978\dots$$

Similarly, from

$$g_{58} = \sqrt{\left(\frac{5 + \sqrt{29}}{2}\right)},$$

we obtain

$$64(g_{58}^{24} + g_{58}^{-24}) = e^{\pi\sqrt{58}} - 24 + 4372e^{-\pi\sqrt{58}} + \dots = 64\left\{\left(\frac{5 + \sqrt{29}}{2}\right)^{12} + \left(\frac{5 - \sqrt{29}}{2}\right)^{12}\right\}.$$

Hence

$$e^{\pi\sqrt{58}} = 24591257751.99999982\dots$$

From:

An Update on Brane Supersymmetry Breaking

J. Mourad and A. Sagnotti - arXiv:1711.11494v1 [hep-th] 30 Nov 2017

From the following vacuum equations:

$$T e^{\gamma_E \phi} = - \frac{\beta_E^{(p)} h^2}{\gamma_E} e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

$$16 k' e^{-2C} = \frac{h^2 \left(p + 1 - \frac{2\beta_E^{(p)}}{\gamma_E} \right) e^{-2(8-p)C + 2\beta_E^{(p)} \phi}}{(7-p)}$$

$$(A')^2 = k e^{-2A} + \frac{h^2}{16(p+1)} \left(7 - p + \frac{2\beta_E^{(p)}}{\gamma_E} \right) e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

we have obtained, from the results almost equals of the equations, putting

$4096 e^{-\pi\sqrt{18}}$ instead of

$$e^{-2(8-p)C + 2\beta_E^{(p)} \phi}$$

a new possible mathematical connection between the two exponentials. Thence, also the values concerning p , C , β_E and ϕ correspond to the exponents of e (i.e. of exp). Thence we obtain for $p = 5$ and $\beta_E = 1/2$:

$$e^{-6C+\phi} = 4096 e^{-\pi\sqrt{18}}$$

Therefore, with respect to the exponentials of the vacuum equations, the Ramanujan's exponential has a coefficient of 4096 which is equal to 642, while $-6C+\phi$ is equal to $-\pi\sqrt{18}$. From this it follows that it is possible to establish mathematically, the dilaton value.

For

$\exp((- \pi \sqrt{18}))$ we obtain:

Input:

$$\exp\left(-\pi \sqrt{18}\right)$$

Exact result:

$$e^{-3 \sqrt{2} \pi}$$

Decimal approximation:

$$1.6272016226072509292942156739117979541838581136954016... \times 10^{-6}$$

$$1.6272016... * 10^{-6}$$

Property:

$$e^{-3 \sqrt{2} \pi} \text{ is a transcendental number}$$

Series representations:

$$e^{-\pi \sqrt{18}} = e^{-\pi \sqrt{17} \sum_{k=0}^{\infty} 17^{-k} \binom{1/2}{k}}$$

$$e^{-\pi \sqrt{18}} = \exp\left(-\pi \sqrt{17} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{17}\right)^k \left(-\frac{1}{2}\right)_k}{k!}\right)$$

$$e^{-\pi \sqrt{18}} = \exp\left(-\frac{\pi \sum_{j=0}^{\infty} \operatorname{Res}_{s=-\frac{1}{2}+j} 17^{-s} \Gamma\left(-\frac{1}{2}-s\right) \Gamma(s)}{2 \sqrt{\pi}}\right)$$

Now, we have the following calculations:

$$e^{-6C+\phi} = 4096e^{-\pi\sqrt{18}}$$

$$e^{-\pi\sqrt{18}} = 1.6272016... * 10^{-6}$$

from which:

$$\frac{1}{4096}e^{-6C+\phi} = 1.6272016... * 10^{-6}$$

$$0.000244140625 e^{-6C+\phi} = e^{-\pi\sqrt{18}} = 1.6272016... * 10^{-6}$$

Now:

$$\ln(e^{-\pi\sqrt{18}}) = -13.328648814475 = -\pi\sqrt{18}$$

And:

$$(1.6272016 * 10^{-6}) * 1 / (0.000244140625)$$

Input interpretation:

$$\frac{1.6272016}{10^6} \times \frac{1}{0.000244140625}$$

Result:

0.0066650177536

0.006665017...

Thence:

$$0.000244140625 e^{-6C+\phi} = e^{-\pi\sqrt{18}}$$

Dividing both sides by 0.000244140625, we obtain:

$$\frac{0.000244140625}{0.000244140625} e^{-6C+\phi} = \frac{1}{0.000244140625} e^{-\pi\sqrt{18}}$$

$$e^{-6C+\phi} = 0.0066650177536$$

$$((((\exp((-Pi*\sqrt{18})))))))*1/0.000244140625$$

Input interpretation:

$$\exp\left(-\pi\sqrt{18}\right)\times\frac{1}{0.000244140625}$$

Result:

0.00666501785...

0.00666501785...

Series representations:

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\pi\sqrt{17} \sum_{k=0}^{\infty} 17^{-k} \left(\frac{1}{2}\right)_k\right)$$

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\pi\sqrt{17} \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{17}\right)^k \left(-\frac{1}{2}\right)_k}{k!}\right)$$

$$\frac{\exp(-\pi\sqrt{18})}{0.000244141} = 4096 \exp\left(-\frac{\pi \sum_{j=0}^{\infty} \operatorname{Res}_{s=-\frac{1}{2}+j} 17^{-s} \Gamma\left(-\frac{1}{2}-s\right) \Gamma(s)}{2\sqrt{\pi}}\right)$$

Now:

$$\begin{aligned}e^{-6\mathcal{C}+\phi} &= 0.0066650177536 \\ \exp\left(-\pi\sqrt{18}\right) \times \frac{1}{0.000244140625} &= \\ e^{-\pi\sqrt{18}} \times \frac{1}{0.000244140625} &= \\ &= 0.00666501785...\end{aligned}$$

From:

$$\ln(0.00666501784619)$$

Input interpretation:

$$\log(0.00666501784619)$$

Result:

$$-5.010882647757...$$

$$-5.010882647757...$$

Alternative representations:

$$\log(0.006665017846190000) = \log_e(0.006665017846190000)$$

$$\log(0.006665017846190000) = \log(a) \log_a(0.006665017846190000)$$

$$\log(0.006665017846190000) = -\mathrm{Li}_1(0.993334982153810000)$$

Series representations:

$$\log(0.006665017846190000) = -\sum_{k=1}^{\infty} \frac{(-1)^k (-0.993334982153810000)^k}{k}$$

$$\log(0.006665017846190000) = 2i\pi \left[\frac{\arg(0.006665017846190000 - x)}{2\pi} \right] + \log(x) - \sum_{k=1}^{\infty} \frac{(-1)^k (0.006665017846190000 - x)^k x^{-k}}{k} \quad \text{for } x < 0$$

$$\log(0.006665017846190000) = \left[\frac{\arg(0.006665017846190000 - z_0)}{2\pi} \right] \log\left(\frac{1}{z_0}\right) + \log(z_0) + \left[\frac{\arg(0.006665017846190000 - z_0)}{2\pi} \right] \log(z_0) - \sum_{k=1}^{\infty} \frac{(-1)^k (0.006665017846190000 - z_0)^k z_0^{-k}}{k}$$

Integral representation:

$$\log(0.006665017846190000) = \int_1^{0.006665017846190000} \frac{1}{t} dt$$

In conclusion:

$$-6C + \phi = -5.010882647757 \dots$$

and for $C = 1$, we obtain:

$$\phi = -5.010882647757 + 6 = \mathbf{0.989117352243} = \phi$$

Note that the values of n_s (spectral index) 0.965, of the average of the Omega mesons Regge slope 0.987428571 and of the dilaton 0.989117352243, are also connected to the following two Rogers-Ramanujan continued fractions:

$$\frac{e^{-\frac{\pi}{5}}}{\sqrt{(\varphi-1)\sqrt{5}} - \varphi + 1} = 1 - \frac{e^{-\pi}}{1 + \frac{e^{-2\pi}}{1 + \frac{e^{-3\pi}}{1 + \frac{e^{-4\pi}}{1 + \dots}}}} \approx 0.9568666373$$

$$\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5} - \varphi + 1} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

(<http://www.bitman.name/math/article/102/109/>)

The mean between the two results of the above Rogers-Ramanujan continued fractions is 0.97798855285, value very near to the ψ Regge slope 0.979:

Ψ		3		$m_c = 1500$		0.979		-0.09
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Also performing the 512th root of the inverse value of the Pion meson rest mass 139.57, we obtain:

$$((1/(139.57)))^{1/512}$$

Input interpretation:

$$\sqrt[512]{\frac{1}{139.57}}$$

Result:

0.990400732708644027550973755713301415460732796178555551684...

0.99040073.... result very near to the dilaton value **0.989117352243 = ϕ** and to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}}}{1 + \sqrt[5]{\sqrt{\phi^5 \sqrt[4]{5^3}} - 1}} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

From

AdS Vacua from Dilaton Tadpoles and Form Fluxes - *J. Mourad and A. Sagnotti*
 - arXiv:1612.08566v2 [hep-th] 22 Feb 2017 - March 27, 2018

We have:

$$e^{2C} = \frac{2\xi e^{\frac{\phi}{2}}}{1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}}$$

$$\frac{h^2}{32} = \frac{\xi^7 e^{4\phi}}{\left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right)^7} \left[\frac{42}{\xi} \left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right) + 5 T e^{2\phi} \right]. \quad (2.7)$$

For

$$T = \frac{16}{\pi^2}$$

$$\xi = 1$$

we obtain:

$$(2 \cdot e^{(0.989117352243/2)}) / (1 + \sqrt{((1 - 1/3 \cdot 16/(\pi)^2 \cdot e^{2 \cdot 0.989117352243})))})$$

Input interpretation:

$$\frac{2 e^{0.989117352243/2}}{1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}}$$

Result:

$$0.83941881822... - \\ 1.4311851867... i$$

Polar coordinates:

$$r = 1.65919106525 \text{ (radius), } \theta = -59.607521917^\circ \text{ (angle)}$$

1.65919106525..... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Series representations:

$$\frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \frac{2 e^{0.4945586761215000}}{1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(\frac{1}{2}\right)_k}$$

$$\frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \frac{2 e^{0.4945586761215000}}{1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16}\right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2}\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!}}$$

$$\frac{2 e^{0.9891173522430000/2}}{1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}}} = \frac{2 e^{0.4945586761215000}}{1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} z_0\right)^k}{k!} z_0^{-k}}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

From

$$\frac{h^2}{32} = \frac{\xi^7 e^{4\phi}}{\left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right)^7} \left[\frac{42}{\xi} \left(1 \pm \sqrt{1 - \frac{\xi T}{3} e^{2\phi}}\right) + 5 T e^{2\phi} \right]$$

We obtain:

$$e^{(4 \times 0.989117352243) / (((1 + \sqrt{1 - 1/3 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}}))^{1/7} [42(1 + \sqrt{1 - 1/3 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}}) + 5 \times 16 / (\pi)^2 \times e^{(2 \times 0.989117352243)}]}$$

Input interpretation:

$$\frac{e^{4 \times 0.989117352243}}{\left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right)^7} \left(42 \left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}}\right) + 5 \times \frac{16}{\pi^2} e^{2 \times 0.989117352243} \right)$$

Result:

50.84107889... -
20.34506335... i

Polar coordinates:

$r = 54.76072411$ (radius), $\theta = -21.80979492^\circ$ (angle)

54.76072411.....

Series representations:

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \right. \\
& \quad \left. \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)_k \right) \right) / \\
& \left(\pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)_k \right) \right)^7 \Bigg) \\
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \right. \\
& \quad \left. \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right) / \\
& \left(\pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^7 \Bigg)
\end{aligned}$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right)^7 = \\
& \left(2 \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \right. \right. \\
& \quad \left. \left. \pi^2 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \\
& \left(\pi^2 \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From which:

$$\begin{aligned}
& e^{(4 \times 0.989117352243)} / (((1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{(2 \times 0.989117352243)}}))^7 \\
& [42(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{(2 \times 0.989117352243)}}) + 5 \times \frac{16}{\pi^2} e^{(2 \times 0.989117352243)}] \times \frac{1}{34}
\end{aligned}$$

Input interpretation:

$$\begin{aligned}
& \frac{e^{4 \times 0.989117352243}}{\left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}} \right)^7} \\
& \left(42 \left(1 + \sqrt{1 - \frac{1}{3} \times \frac{16}{\pi^2} e^{2 \times 0.989117352243}} \right) + 5 \times \frac{16}{\pi^2} e^{2 \times 0.989117352243} \right) \times \frac{1}{34}
\end{aligned}$$

Result:

1.495325850... -
0.5983842161... *i*

Polar coordinates:

$r = 1.610609533$ (radius), $\theta = -21.80979492^\circ$ (angle)

1.610609533.... result that is a good approximation to the value of the golden ratio
1.618033988749...

Series representations:

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 = \\
& \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \\
& \quad \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)_k \right) / \\
& \quad \left(17 \pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \left(\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(\frac{1}{2} \right)_k \right) \right)^7 \\
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 = \\
& \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \pi^2 \right. \\
& \quad \left. \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) / \\
& \quad \left(17 \pi^2 \left(1 + \sqrt{-\frac{16 e^{1.978234704486000}}{3 \pi^2}} \sum_{k=0}^{\infty} \frac{\left(-\frac{3}{16} \right)^k \left(-\frac{e^{1.978234704486000}}{\pi^2} \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)^7
\end{aligned}$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) + \frac{5 \times 16 e^{2 \times 0.9891173522430000}}{\pi^2} \right) \right. \\
& \quad \left. e^{4 \times 0.9891173522430000} \right) / \left(34 \left(1 + \sqrt{1 - \frac{16 e^{2 \times 0.9891173522430000}}{3 \pi^2}} \right) \right)^7 = \\
& \left(40 e^{5.934704113458000} + 21 e^{3.956469408972000} \pi^2 + 21 e^{3.956469408972000} \right. \\
& \quad \left. \pi^2 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) / \\
& \left(17 \pi^2 \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 - \frac{16 e^{1.978234704486000}}{3 \pi^2} - z_0\right)^k z_0^{-k}}{k!} \right) \right)^7 \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

Now, we have:

$$e^{2C} = \frac{2\xi e^{-\frac{\phi}{2}}}{1 + \sqrt{1 + \frac{\xi\Lambda}{3} e^{2\phi}}}, \quad (2.9)$$

$$\frac{h^2}{32} = \frac{e^{-4\phi}}{\left[1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right]^7} \left[42 \left(1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}}\right) - 13\Lambda e^{2\phi}\right]. \quad (2.10)$$

For:

$$\xi = 1$$

$$\Lambda \simeq \frac{4\pi^2}{25}$$

$$\phi = 0.989117352243$$

From

$$e^{2C} = \frac{2\xi e^{-\frac{\phi}{2}}}{1 + \sqrt{1 + \frac{\xi\Lambda}{3} e^{2\phi}}},$$

we obtain:

$$\frac{((2 * e^{(-0.989117352243/2)}))}{((((1 + \sqrt{(1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)})))))}$$

Input interpretation:

$$\frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \cdot 0.989117352243}}}$$

Result:

0.382082347529...

0.382082347529....

Series representations:

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = 2 / \left(e^{0.4945586761215000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right)_k \right) \right)$$

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = 2 / \left(e^{0.4945586761215000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2} \right)_k}{k!} \right) \right)$$

$$\frac{2 e^{-0.9891173522430000/2}}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} = \frac{2}{e^{0.4945586761215000} \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} z_0\right)^k z_0^{-k}}{k!} \right)}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

From which:

$$1 + 1 / (((4((2 * e^{(-0.989117352243/2)})) / (((1 + \sqrt{((1 + 1/3 * (4 \pi^2)/25 * e^{(2 * 0.989117352243))}))))))))))$$

Input interpretation:

$$1 + \frac{1}{4 \times \frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}}}}$$

Result:

1.65430921270...

1.6543092..... We note that, the result 1.6543092... is very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Indeed:

$$G_{505} = P^{-1/4}Q^{1/6} = (\sqrt{5} + 2)^{1/2} \left(\frac{\sqrt{5} + 1}{2} \right)^{1/4} (\sqrt{101} + 10)^{1/4} \\ \times \left((130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} \right)^{1/6}.$$

Thus, it remains to show that

$$(130\sqrt{5} + 29\sqrt{101}) + \sqrt{169440 + 7540\sqrt{505}} = \left(\sqrt{\frac{113 + 5\sqrt{505}}{8}} + \sqrt{\frac{105 + 5\sqrt{505}}{8}} \right)^3,$$

which is straightforward. $\square$

$$\sqrt[14]{\left(\sqrt{\frac{113+5\sqrt{505}}{8}} + \sqrt{\frac{105+5\sqrt{505}}{8}} \right)^3} = 1,65578 \dots$$

Series representations:

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = \\ \frac{1 + \sqrt{1 + \frac{(4\pi^2)e^{2 \times 0.9891173522430000}}{3 \times 25}}}{1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}}} \\ \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)_k$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = \\ \frac{1 + \sqrt{1 + \frac{(4\pi^2)e^{2 \times 0.9891173522430000}}{3 \times 25}}}{1 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}}} \\ \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} = 1 + \frac{e^{0.4945586761215000}}{8} +$$

$$\frac{1}{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}} \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2} \right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0 \right)^k z_0^{-k}}{k!}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

And from

$$\frac{h^2}{32} = \frac{e^{-4\phi}}{\left[1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}} \right]^7} \left[42 \left(1 + \sqrt{1 + \frac{\Lambda}{3} e^{2\phi}} \right) - 13 \Lambda e^{2\phi} \right].$$

we obtain:

$$e^{(-4 \times 0.989117352243) / [1 + \sqrt{((1 + 1/3 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243))})}]^7 \times [42(1 + \sqrt{((1 + 1/3 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243))})}) - 13 \times (4\pi^2)/25 \times e^{(2 \times 0.989117352243)}]}$$

Input interpretation:

$$\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7 \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right)}$$

Result:

-0.034547055658...

-0.034547055658...

Series representations:

$$\left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \right. \\ \left. - \left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\ \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\ \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right) \right) / \left(25 e^{5.934704113458000} \right. \\ \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right)^7 \right) \right)$$

$$\begin{aligned}
& \left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right. \right. \\
& \quad \left. \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \right. \\
& \quad - \left(\left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
& \quad \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^7 \right) \right) \\
& \left(\left(42 \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \frac{1}{25} (4\pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right. \\
& \quad \left. e^{-4 \times 0.9891173522430000} \right) / \left(1 + \sqrt{1 + \frac{(4\pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& \quad - \left(\left(42 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \right. \\
& \quad \left. \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \left(25 \right. \\
& \quad \left. e^{5.934704113458000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From which:

$$\begin{aligned}
& 47 * 1 / (((-1 / (((((e^{(-4 * 0.989117352243)} / \\
& [1 + \sqrt{((1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}))])^7 * \\
& [42(1 + \sqrt{((1 + 1/3 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}))}) - \\
& 13 * (4\pi^2)/25 * e^{(2 * 0.989117352243)}])))))))
\end{aligned}$$

Input interpretation:

$$47 \left(- \left(1 / 1 / \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \right. \right. \\ \left. \left. \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - \right. \right. \right. \\ \left. \left. \left. 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right) \right) \right)$$

Result:

1.6237116159...

1.6237116159.... result that is an approximation to the value of the golden ratio
1.618033988749...

Series representations:

$$- \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\ \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\ \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\ \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \\ \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\ \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right) / \left(25 e^{5.934704113458000} \right. \\ \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right)^7$$

$$\begin{aligned}
& - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \\
& \quad 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \\
& \quad \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k \left(e^{1.978234704486000} \pi^2\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \Bigg) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k \left(e^{1.978234704486000} \pi^2\right)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^7 \right) \\
& - \left(47 / 1 / \left(e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& \left(1974 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \\
& \quad \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \Bigg) / \left(25 e^{5.934704113458000} \right. \\
& \quad \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

And again:

$$32((((e^{(-4*0.989117352243)} / [1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)})})}]^7 * [42(1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243)})})})-13*(4\pi^2)/25*e^{(2*0.989117352243)})])])])$$

Input interpretation:

$$32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right)$$

Result:

-1.1055057810...

-1.1055057810....

We note that the result -1.1055057810.... is very near to the value of Cosmological Constant, less 10^{-52} , thence 1.1056, with minus sign

Series representations:

$$\begin{aligned}
 & \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
 & \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
 & \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
 & - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
 & \quad \left. \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\
 & \quad \left. \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
 & \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \binom{\frac{1}{2}}{k} \right)^7 \right) \right)
 \end{aligned}$$

$$\begin{aligned}
& \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - \right. \right. \right. \\
& \quad \left. \left. \left. 25 e^{1.978234704486000} \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
& \quad \left. \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) \right)^7 \right) \right)
\end{aligned}$$

$$\begin{aligned}
& \left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 = \\
& - \left(\left(1344 \left(-25 e^{1.978234704486000} + 52 e^{3.956469408972000} \pi^2 - 25 e^{1.978234704486000} \right. \right. \right. \\
& \quad \left. \left. \left. \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) \right) / \left(25 e^{5.934704113458000} \right. \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

And:

$$- [32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \\ \left. [42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}] \right)^5$$

Input interpretation:

$$- \left[32 \left(\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right)^7} \right. \right. \\ \left. \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}} \right) - \right. \right. \\ \left. \left. \left. 13 \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243} \right) \right)^5 \right]$$

Result:

1.651220569...

1.651220569.... result very near to the 14th root of the following Ramanujan's class invariant $Q = (G_{505}/G_{101/5})^3 = 1164.2696$ i.e. 1.65578...

Series representations:

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Big)^5 = \\
& \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)^5 \right) \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)^{35} \right) \right)
\end{aligned}$$

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^5 \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^{35} \right)
\end{aligned}$$

$$\begin{aligned}
& - \left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right) - \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg)^5 = \\
& \left(4385270057140224 \left(-25 + 52 e^{1.978234704486000} \pi^2 - \right. \right. \\
& \quad \left. \left. 25 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^5 \right) / \\
& \quad \left(9765625 e^{19.78234704486000} \right. \\
& \quad \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^{35} \right)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

We obtain also:

$$-\left[32\left(\frac{e^{-4 \times 0.989117352243}}{\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)^7}\right)}\right)^7 \times \left[42\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)}\right)-13\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}\right)\right]\right)^{1 / 2}$$

Input interpretation:

$$-\sqrt{\left(32\left(\frac{e^{-4 \times 0.989117352243}}{\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)^7}\right)}\right)^7 \times \left[42\left(1+\sqrt{1+\frac{1}{3}\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}}\right)}\right)-13\left(\frac{1}{25}\left(4 \pi^2\right) e^{2 \times 0.989117352243}\right)\right]\right)^{1 / 2}}$$

Result:

$$-0.10514303501 \dots i$$

Polar coordinates:

$$r = 1.05143035007 \text{ (radius), } \theta = -90^\circ \text{ (angle)}$$

$$1.05143035007$$

Series representations:

$$\begin{aligned}
 & - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
 & \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \right. \\
 & \quad \left. \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \right) = -\frac{8}{5} \sqrt{21} \\
 & \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
 & \quad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right) \right) / \left(e^{3.956469408972000} \right. \\
 & \quad \left. \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(\frac{1}{2} \right) \right)^7 \right) \right)
 \end{aligned}$$

$$\begin{aligned}
& - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = -\frac{8}{5} \sqrt{21} \\
& \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \\
& \quad \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right) / \right. \\
& \quad \left. \left(e^{3.956469408972000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\
& \quad \left. \left. \left. \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4}\right)^k (e^{1.978234704486000} \pi^2)^{-k} \left(-\frac{1}{2}\right)_k}{k!} \right)^7 \right) \right) \Bigg) \\
& - \sqrt{\left(\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} - \right. \right. \right. \right. \\
& \quad \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) / \\
& \quad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) = \\
& -\frac{8}{5} \sqrt{21} \sqrt{\left(\left(25 - 52 e^{1.978234704486000} \pi^2 + \right. \right. \\
& \quad \left. \left. 25 \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right) / \right. \\
& \quad \left(e^{3.956469408972000} \right. \\
& \quad \left. \left. \left(1 + \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \right)^7 \right) \right) \Bigg)
\end{aligned}$$

for (not $(z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0)$)

$$\frac{e^{-\frac{\pi}{5}}}{\sqrt{(\varphi-1)\sqrt{5}}-\varphi+1} = 1 - \frac{e^{-\pi}}{1 + \frac{e^{-2\pi}}{1 + \frac{e^{-3\pi}}{1 + \frac{e^{-4\pi}}{1 + \dots}}}} \approx 0.9568666373$$

Series representations:

$$\begin{aligned} & - \left[1 / \left(\sqrt{\left(32 e^{-4 \times 0.9891173522430000} \left(42 \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right. \right. \right. \right. \right. \right. \right. \\ & \qquad \qquad \qquad \left. \left. \left. \left. \left. \frac{1}{25} (4 \pi^2) 13 e^{2 \times 0.9891173522430000} \right) \right) \right) \right) \right] / \\ & \qquad \qquad \qquad \left(1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}} \right)^7 \Bigg) \Bigg) = \\ & - \left[5 / \left(8 \sqrt{21} \sqrt{\left(25 - 52 e^{1.978234704486000} \pi^2 + 25 \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right. \right. \right. \\ & \qquad \qquad \qquad \left. \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right) \right] / \\ & \qquad \qquad \qquad \left(e^{3.956469408972000} \left(1 + \sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75}} \right) \right. \\ & \qquad \qquad \qquad \left. \left. \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right) \right) \right)^7 \Bigg) \Bigg) \Bigg) \Bigg) \Bigg) \end{aligned}$$

for (not ($z_0 \in \mathbb{R}$ and $-\infty < z_0 \leq 0$))

From the previous expression

$$\frac{e^{-4 \times 0.989117352243}}{\left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}}\right)^7 \left(42 \left(1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}}\right) - 13 \left(\frac{1}{25} (4 \pi^2)\right) e^{2 \times 0.989117352243}\right)}$$

$$= -0.034547055658...$$

we have also:

$$1+1/(((4((2*e^{(-0.989117352243/2)})) / (((1+\sqrt{((1+1/3*(4\pi^2)/25*e^{(2*0.989117352243))}})))))) + (-0.034547055658)$$

Input interpretation:

$$1 + \frac{1}{4 \times \frac{2 e^{-0.989117352243/2}}{1 + \sqrt{1 + \frac{1}{3} \left(\frac{1}{25} (4 \pi^2) \right) e^{2 \times 0.989117352243}}}} - 0.034547055658$$

Result:

1.61976215705...

1.61976215705..... result that is a very good approximation to the value of the golden ratio 1.618033988749...

Series representations:

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 =$$

$$\frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000}}$$

$$\sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75} \sum_{k=0}^{\infty} \left(\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(\frac{1}{2} \right)_k}$$

$$1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 =$$

$$\frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} + \frac{1}{8} e^{0.4945586761215000}}$$

$$\sqrt{\frac{4 e^{1.978234704486000} \pi^2}{75} \sum_{k=0}^{\infty} \frac{\left(-\frac{75}{4} \right)^k \left(e^{1.978234704486000} \pi^2 \right)^{-k} \left(-\frac{1}{2} \right)_k}{k!}}$$

$$\begin{aligned}
& 1 + \frac{1}{4 \left(2 e^{-0.9891173522430000/2} \right)} - 0.0345470556580000 = \\
& \frac{1 + \sqrt{1 + \frac{(4 \pi^2) e^{2 \times 0.9891173522430000}}{3 \times 25}}}{0.9654529443420000 + \frac{e^{0.4945586761215000}}{8} +} \\
& \frac{1}{8} e^{0.4945586761215000} \sqrt{z_0} \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{1}{2}\right)_k \left(1 + \frac{4 e^{1.978234704486000} \pi^2}{75} - z_0\right)^k z_0^{-k}}{k!} \\
& \text{for (not } (z_0 \in \mathbb{R} \text{ and } -\infty < z_0 \leq 0))
\end{aligned}$$

From

Properties of Nilpotent Supergravity

E. Dudas, S. Ferrara, A. Kehagias and A. Sagnotti - arXiv:1507.07842v2 [hep-th] 14 Sep 2015

We have that:

Cosmological inflation with a tiny tensor-to-scalar ratio r , consistently with PLANCK data, may also be described within the present framework, for instance choosing

$$\alpha(\Phi) = i M \left(\Phi + b \Phi e^{ik\Phi} \right). \quad (4.35)$$

This potential bears some similarities with the Kähler moduli inflation of [32] and with the poly-instanton inflation of [33]. One can verify that $\chi = 0$ solves the field equations, and that the potential along the $\chi = 0$ trajectory is now

$$V = \frac{M^2}{3} \left(1 - a \phi e^{-\gamma \phi} \right)^2. \quad (4.36)$$

We analyzing the following equation:

$$V = \frac{M^2}{3} \left(1 - a \phi e^{-\gamma \phi} \right)^2 .$$

$$\phi = \varphi - \frac{\sqrt{6}}{k} ,$$

$$a = \frac{b\gamma}{e} < 0 , \quad \gamma = \frac{k}{\sqrt{6}} < 0 .$$

We have:

$$(M^2)/3*[1-(b/\text{euler number} * k/\sqrt{6}) * (\varphi - \sqrt{6}/k) * \exp(-(k/\sqrt{6})(\varphi - \sqrt{6}/k)))]^2$$

i.e.

$$V = (M^2)/3*[1-(b/\text{euler number} * k/\sqrt{6}) * (\varphi - \sqrt{6}/k) * \exp(-(k/\sqrt{6})(\varphi - \sqrt{6}/k)))]^2$$

For $k = 2$ and $\varphi = 0.9991104684$, that is the value of the scalar field that is equal to the value of the following Rogers-Ramanujan continued fraction:

$$\frac{\frac{e^{-\frac{\pi}{\sqrt{5}}}}{\sqrt{5}}}{1 + \sqrt[5]{\sqrt{\varphi^5 \sqrt[4]{5^3}} - 1} - \varphi + 1} = 1 - \frac{e^{-\pi\sqrt{5}}}{1 + \frac{e^{-2\pi\sqrt{5}}}{1 + \frac{e^{-3\pi\sqrt{5}}}{1 + \frac{e^{-4\pi\sqrt{5}}}{1 + \dots}}}} \approx 0.9991104684$$

we obtain:

$$V = (M^2)/3*[1-(b/\text{euler number} * 2/\sqrt{6}) * (0.9991104684 - \sqrt{6}/2) * \exp(-(2/\sqrt{6})(0.9991104684 - \sqrt{6}/2)))]^2$$

Input interpretation:

$$V = \frac{M^2}{3} \left(1 - \left(\frac{b}{e} \times \frac{2}{\sqrt{6}} \right) \left(0.9991104684 - \frac{\sqrt{6}}{2} \right) \exp \left(- \frac{2}{\sqrt{6}} \left(0.9991104684 - \frac{\sqrt{6}}{2} \right) \right) \right)^2$$

Result:

$$V = \frac{1}{3} (0.0814845 b + 1)^2 M^2$$

Solutions:

$$b = \frac{225.913 \left(-0.054323 M^2 \pm 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} \quad (M \neq 0)$$

Alternate forms:

$$V = 0.00221324 (b + 12.2723)^2 M^2$$

$$V = 0.00221324 (b^2 M^2 + 24.5445 b M^2 + 150.609 M^2)$$

$$-0.00221324 b^2 M^2 - 0.054323 b M^2 - \frac{M^2}{3} + V = 0$$

Expanded form:

$$V = 0.00221324 b^2 M^2 + 0.054323 b M^2 + \frac{M^2}{3}$$

Alternate form assuming b, M, and V are positive:

$$V = 0.00221324 (b + 12.2723)^2 M^2$$

Alternate form assuming b, M, and V are real:

$$V = 0.00221324 b^2 M^2 + 0.054323 b M^2 + 0.333333 M^2 + 0$$

Derivative:

$$\frac{\partial}{\partial b} \left(\frac{1}{3} (0.0814845 b + 1)^2 M^2 \right) = 0.054323 (0.0814845 b + 1) M^2$$

Implicit derivatives:

$$\frac{\partial b(M, V)}{\partial V} = \frac{154317775011120075}{36961748(226802245 + 18480874b)M^2}$$

$$\frac{\partial b(M, V)}{\partial M} = - \frac{\frac{226802245}{18480874} + b}{M}$$

$$\frac{\partial M(b, V)}{\partial V} = \frac{154317775011120075}{2(226802245 + 18480874b)^2 M}$$

$$\frac{\partial M(b, V)}{\partial b} = - \frac{18480874 M}{226802245 + 18480874 b}$$

$$\frac{\partial V(b, M)}{\partial M} = \frac{2(226802245 + 18480874b)^2 M}{154317775011120075}$$

$$\frac{\partial V(b, M)}{\partial b} = \frac{36961748 (226802245 + 18480874 b) M^2}{154317775011120075}$$

Global minimum:

$$\min\left\{\frac{1}{3} (0.0814845 b + 1)^2 M^2\right\} = 0 \text{ at } (b, M) = (-16, 0)$$

Global minima:

$$\min\left\{\frac{1}{3} M^2 \left(1 - \frac{(b+2) \left(0.9991104684 - \frac{\sqrt{6}}{2}\right) \exp\left(-\frac{2\left(0.9991104684 - \frac{\sqrt{6}}{2}\right)}{\sqrt{6}}\right)}{e\sqrt{6}}\right)^2\right\} = 0$$

for $b = -\frac{226802245}{18480874}$

$$\min\left\{\frac{1}{3} M^2 \left(1 - \frac{(b+2) \left(0.9991104684 - \frac{\sqrt{6}}{2}\right) \exp\left(-\frac{2\left(0.9991104684 - \frac{\sqrt{6}}{2}\right)}{\sqrt{6}}\right)}{e\sqrt{6}}\right)^2\right\} = 0$$

for $M = 0$

From:

$$b = \frac{225.913 \left(-0.054323 M^2 \pm 6.58545 \times 10^{-10} \sqrt{M^4}\right)}{M^2} \quad (M \neq 0)$$

we obtain

$$(225.913 (-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4}))/M^2$$

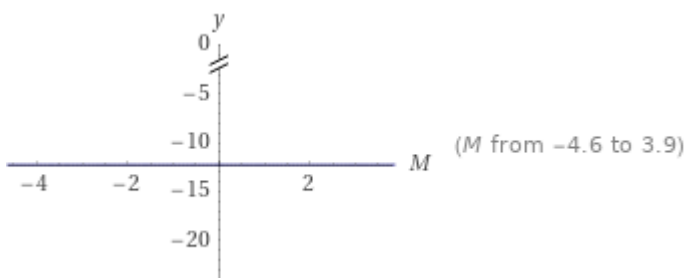
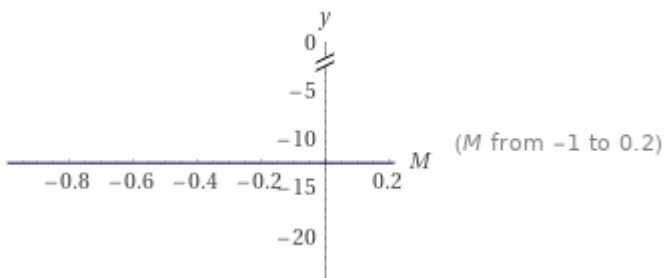
Input interpretation:

$$\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2}$$

Result:

$$\frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2}$$

Plots:



Alternate form assuming M is real:

$$-12.2723$$

-12.2723 result very near to the black hole entropy value $12.1904 = \ln(196884)$

Alternate forms:

$$- \frac{12.2723 \left(M^2 - 1.21228 \times 10^{-8} \sqrt{M^4} \right)}{M^2}$$

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4} - 12.2723 M^2}{M^2}$$

Expanded form:

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M^2} - 12.2723$$

Property as a function:**Parity**

even

Series expansion at $M = 0$:

$$\left(\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M^2} - 12.2723 \right) + O(M^6)$$

(generalized Puiseux series)

Series expansion at $M = \infty$:

$$-12.2723$$

Derivative:

$$\frac{d}{dM} \left(\frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right) = \frac{3.55271 \times 10^{-15}}{M}$$

Indefinite integral:

$$\int \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} dM =$$

$$\frac{1.48774 \times 10^{-7} \sqrt{M^4}}{M} - 12.2723 M + \text{constant}$$

Global maximum:

$$\max \left\{ \frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right\} =$$

$$- \frac{140\,119\,826\,723\,990\,341\,497\,649}{11\,417\,594\,849\,251\,000\,000\,000} \text{ at } M = -1$$

Global minimum:

$$\min \left\{ \frac{225.913 \left(6.58545 \times 10^{-10} \sqrt{M^4} - 0.054323 M^2 \right)}{M^2} \right\} =$$

$$- \frac{140\,119\,826\,723\,990\,341\,497\,649}{11\,417\,594\,849\,251\,000\,000\,000} \text{ at } M = -1$$

Limit:

$$\lim_{M \rightarrow \pm\infty} \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} = -12.2723$$

Definite integral after subtraction of diverging parts:

$$\int_0^\infty \left(\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} - -12.2723 \right) dM = 0$$

From b that is equal to

$$\frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2}$$

from:

Result:

$$V = \frac{1}{3} (0.0814845 b + 1)^2 M^2$$

we obtain:

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4}))/M^2) + 1)^2 M^2$$

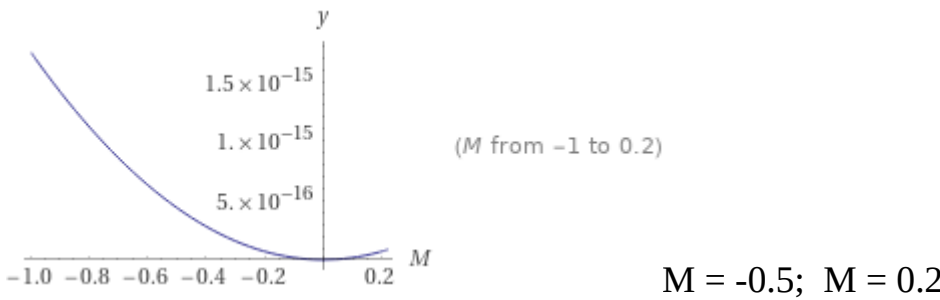
Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

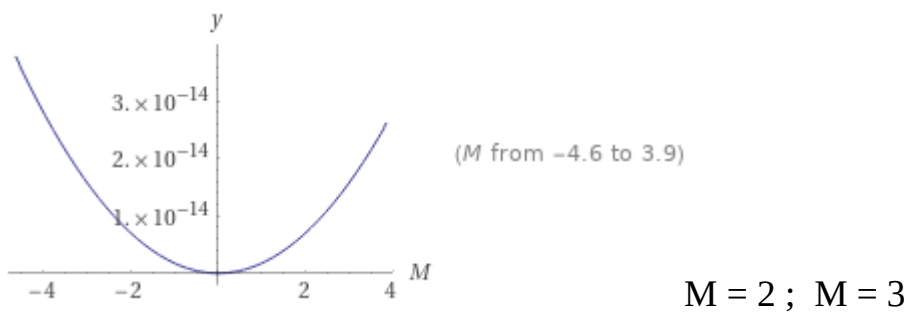
Result:

0

Plots: (possible mathematical connection with an open string)



(possible mathematical connection with an open string)



Root:

$M = 0$

Property as a function:

Parity

even

Series expansion at $M = 0$:

$$O(M^{62194})$$

(Taylor series)

Series expansion at $M = \infty$:

$$1.75541 \times 10^{-15} M^2 + O\left(\left(\frac{1}{M}\right)^{62194}\right)$$

(Taylor series)

Definite integral after subtraction of diverging parts:

$$\int_0^\infty \left(\frac{1}{3} M^2 \left(1 + \frac{18.4084 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} \right)^2 - 1.75541 \times 10^{-15} M^2 \right) dM = 0$$

For $M = -0.5$, we obtain:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right) M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 (-0.5)^2 + 6.58545 \times 10^{-10} \sqrt{(-0.5)^4}))) / (-0.5)^2 + 1)^2 * (-0.5^2)$$

Input interpretation:

$$\frac{1}{3} \left(\frac{0.0814845 \times \frac{225.913 \left(-0.054323 (-0.5)^2 + 6.58545 \times 10^{-10} \sqrt{(-0.5)^4} \right)}{(-0.5)^2}}{(-0.5^2)} + 1 \right)^2$$

Result:

$$-4.38851344947464545348970783378088020833333333333333333\ldots \times 10^{-16}$$

$$-4.38851344947 \times 10^{-16}$$

For $M = 0.2$:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 \cdot 0.2^2 + 6.58545 \times 10^{-10} \sqrt{0.2^4}))) / 0.2^2 + 1)^2 \cdot 0.2^2$$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 \times 0.2^2 + 6.58545 \times 10^{-10} \sqrt{0.2^4} \right)}{0.2^2} + 1 \right)^2 \times 0.2^2$$

Result:

$$7.0216215191594327255835325340494083333333333333333333333 \times 10^{-17}$$

$$7.021621519159 \times 10^{-17}$$

For $M = 3$:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 \cdot 3^2 + 6.58545 \times 10^{-10} \sqrt{3^4}))) / 3^2 + 1)^2 \cdot 3^2$$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 \times 3^2 + 6.58545 \times 10^{-10} \sqrt{3^4} \right)}{3^2} + 1 \right)^2 \times 3^2$$

Result:

$$1.579864841810872363256294820161116875 \times 10^{-14}$$

$$1.57986484181 \times 10^{-14}$$

For $M = 2$:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 M^2 + 6.58545 \times 10^{-10} \sqrt{M^4} \right)}{M^2} + 1 \right)^2 M^2$$

$$\frac{1}{3} (0.0814845 ((225.913 (-0.054323 \cdot 2^2 + 6.58545 \times 10^{-10} \sqrt{2^4}))) / 2^2 + 1)^2 \cdot 2^2$$

Input interpretation:

$$\frac{1}{3} \left(0.0814845 \times \frac{225.913 \left(-0.054323 \times 2^2 + 6.58545 \times 10^{-10} \sqrt{2^4} \right)}{2^2} + 1 \right)^2 \times 2^2$$

Result:

$$7.0216215191594327255835325340494083333333333333333333333... \times 10^{-15}$$

$$7.021621519 \times 10^{-15}$$

From the four results

7.021621519*10⁻¹⁵ ; 1.57986484181*10⁻¹⁴ ; 7.021621519159*10⁻¹⁷ ;
-4.38851344947*10⁻¹⁶

we obtain, after some calculations:

$$\sqrt{[1/(2\pi)(7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16})]}$$

Input interpretation:

$$\sqrt{\left(\frac{1}{2\pi} \left(7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16}\right)\right)}$$

Result:

$$5.9776991059... \times 10^{-8}$$

5.9776991059*10⁻⁸ result very near to the Planck's electric flow 5.975498×10^{-8} that is equal to the following formula:

$$\phi_P^E = \mathbf{E}_P l_P^2 = \phi_P l_P = \sqrt{\frac{\hbar c}{\epsilon_0}}$$

We note that:

$$1/55 * (((((1/[(7.021621519*10^{-15} + 1.57986484181*10^{-14} + 7.021621519*10^{-17} - 4.38851344947*10^{-16})]))^{1/7} - ((\log^{5/8}(2))/(2 \cdot 2^{1/8} \cdot 3^{1/4} \cdot e \log^{3/2}(3))))))$$

Input interpretation:

$$\frac{1}{55} \left(\left(1 / (7.021621519 \times 10^{-15} + 1.57986484181 \times 10^{-14} + 7.021621519 \times 10^{-17} - 4.38851344947 \times 10^{-16}) \right)^{(1/7)} - \frac{\log^{5/8}(2)}{2 \sqrt[8]{2} \sqrt[4]{3} e \log^{3/2}(3)} \right)$$

$\log(x)$ is the natural logarithm

Result:

$$1.6181818182...$$

1.6181818182... result that is a very good approximation to the value of the golden ratio 1.618033988749...

From the Planck units:

Planck Length

$$l_P = \sqrt{\frac{4\pi\hbar G}{c^3}}$$

$5.729475 \cdot 10^{-35}$ Lorentz-Heaviside value

Planck's Electric field strength

$$\mathbf{E}_P = \frac{F_P}{q_P} = \sqrt{\frac{c^7}{16\pi^2 \epsilon_0 \hbar G^2}}$$

$1.820306 \cdot 10^{61}$ V*m Lorentz-Heaviside value

Planck's Electric flux

$$\phi_P^E = \mathbf{E}_P l_P^2 = \phi_P l_P = \sqrt{\frac{\hbar c}{\epsilon_0}}$$

$5.975498 \cdot 10^{-8}$ V*m Lorentz-Heaviside value

Planck's Electric potential

$$\phi_P = V_P = \frac{E_P}{q_P} = \sqrt{\frac{c^4}{4\pi\epsilon_0 G}}$$

$1.042940 \cdot 10^{27}$ V Lorentz-Heaviside value

Relationship between Planck's Electric Flux and Planck's Electric Potential

$$E_P * I_P = (1.820306 * 10^{61}) * 5.729475 * 10^{-35}$$

Input interpretation:

$$\frac{(1.820306 \times 10^{61}) \times 5.729475}{10^{35}}$$

Result:

1 042 939 771 935 000 000 000 000 000

Scientific notation:

$$1.042939771935 \times 10^{27}$$

$$1.042939771935 * 10^{27} \approx 1.042940 * 10^{27}$$

Or:

$$E_P * I_P^2 / I_P = (5.975498 * 10^{-8}) * 1 / (5.729475 * 10^{-35})$$

Input interpretation:

$$5.975498 \times 10^{-8} \times \frac{1}{\frac{5.729475}{10^{35}}}$$

Result:

1.04293988541707573556041347592929544155441816222254220500133... × 10²⁷

$$1.042939885417 * 10^{27} \approx 1.042940 * 10^{27}$$

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